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Inflation Classical Marx Growth + Inflation

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Inflation

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then obviously we would not be able to make any judgement about how to proceed. In addition, the emphasis on the efficiency calculus, I think, is just on that particular mode of evaluation of what are admittedly complex and fundamentally uncertain phenomena. Political practice, and scientific practice, in so far as it is different from political practice and human life itself, is based on evaluations of outcomes. We cannot exist without doing that, and so whether or not we do it perfectly, we do it. And the best thing we can try to do is to try to improve how we do it and confront our mistakes, theorizing with the empirical and historical evidence and learn from our mistakes. And I take that to be one of the themes of the chapter – and I hope a correct reading.

4 Explaining Inflation and Unemployment: An Alternative to Neoliberal Economic Theory

Anwar M. Shaikh

INTRODUCTION

For much of the post-war period, the problems of inflation and unemployment have been central to economic and political agendas. And in this domain Neoliberal economics has come to dominate both modern macroeconomic theory and policy.

Capitalism has undergone a world-wide economic crisis in the last two decades. Its response has been a series of attacks on labour and its supporting institutions, widespread business failures and bankruptcies, a dizzying spiral of concentration and centralization and an urgent drive to make available new markets and new resource areas to the unchecked power of the dominant world capitals (Shaikh 1987). Neoliberal economic *policy* arose out of the need to support and coordinate these characteristic responses of the capitalist class.

But Neoliberal economic *theory* came to the fore because Keynesian theory was unable to provide an adequate explanation for the 'stagflation' deriving from the economic crisis. This is a particular irony, given that Keynesian economic theory itself rose to power 30 years earlier because the neoclassical economic theory which underpins neoliberal economics was itself unable to explain the huge and persistent unemployment of the last great depression.

Modern heterodox macroeconomics finds itself caught in this conflict, because by the 1970s much of it had come to be subsumed under Keynesian concerns. Thus both radical and postkeynesian economics typically begin from some version of Keynesian or Kaleckian effective demand theory: a static equilibrium framework in which markup pricing insulates prices from demand, thereby shifting the adjustment process onto output and employment – at least until the vicinity of full employment. Within this framework, the obstacle to full employment with stable prices is generally political, not economic, rooted in the welter of conflicting interests arising out of the tri-cornered tug-of-war between capital, labour and the nation-state. The marxian wing of this tradition differs only in its somewhat greater emphasis on monopoly power and on the potential problems associated with 'full employment' (Kalecki 1968).

Neoclassical economic theory has no such problem, for it assumes that the capitalist system delivers full employment automatically and efficiently. In its basic form, inflation arises when an increase in the money supply stimulates aggregate demand in the face of a full-employment-constrained aggregate supply. More recent versions incorporating concepts such as the natural rate of unemployment are merely refinements of this basic argument. Here too, as in Keynesian–Kaleckian theory, inflation is expected to arise in the vicinity of full employment.

In contrast to these familiar perspectives, I would like to present a classical marxian explanation of inflation and its relationship (or lack thereof) to unemployment. Broadly speaking, a classical marxian framework considers economic growth to be the normal state of a capitalist economy, driven by the ceaseless attempts of each individual capital to constantly (self-) expand. Since each capital operates individually, without any direct regard for its place in the overall social division of labour, the interaction of these individual units produces an intrinsically turbulent process: the imagined division of labour created by the expectations of individual capitals is constantly confronted by an actual division of labour created by their own mutual actions, and the discrepancies react back on both expectations and actions, creating fresh discrepancies, and so on. This inherently turbulent process is precisely what neoclassical economics tries to cover up through its recourse to perfect competition and general equilibrium. But in fact, disequilibrium is always the existing condition, and it is precisely through offsetting phases of overshooting and undershooting that the inner tendencies are realized. From this point of view, balance conditions of various sort (demand–supply, output–capacity, sectoral growth, and so on) represent the inner forces which impose a hidden order on the outer disorder: order-in-and-through-disorder, an old concept in Marx which has finally been given legitimacy via non-linear dynamics.

In my own work, I have tried to show that this approach can be formalized so as to provide an *integrated* dynamic non-equilibrium framework for the analysis of endogenous growth, endogenous money and endogenous cycles (Shaikh 1989, 1991, 1992). Building upon Goodwin's classic work (Goodwin 1967), such a framework can be extended to incorporate an endogenous theory of persistent unemployment rooted in competition itself. This is what Marx calls a 'reserve army of labour', which we might today call an 'intrinsic rate of unemployment' to distinguish it from the pernicious Neoliberal idea of a 'natural rate of unemployment'. The former concept is rooted in the notion that the system is behaving perfectly well when it creates and maintains a pool of involuntarily unemployed people at the disposal of capital; the latter claims that it is imperfections in the system which give rise to voluntary employment – that is, to abstentions from work (Friedman 1968).

In the present chapter, I would like to address the other great problem: the question of inflation and its links, if any, to unemployment. I will trace the treatment of these issues in orthodox theories, as they arose historically in the face of challenges from historical events. Then I will outline an alternate approach to the question of inflation, and illustrate it with data for an average of the main OECD countries, and for the United States in particular.

UNEMPLOYMENT AND INFLATION IN THEORY AND HISTORY

Modern macroeconomics has its origins in the turmoil of the great depression of the 1930s. While the prevailing economic theory continued to insist that capitalism was intrinsically efficient, self-regulating, and automatically able to offer employment to all who desired it, economic reality told a different story. Widespread business failures, massive unemployment, generalized social misery – these were the social and historical facts of the day. It is in this context that Keynes' *General Theory* (Keynes 1936, 1964) stepped forward to provide an explanation for persistent unemployment, as well as a prescription for its cure. The familiar income–expenditure model derived from this approach was to dominate both macroeconomic theory and policy for a third of a century in most of the advanced capitalist world. It was systematic, quantifiable, flexible in its application and easily adapted to fiscal policy. The model is driven by exogenous components of aggregate demand, under the assumption that there are unemployed resources, most notably labour. A rise in the exogenous demand component stimulates output and employment, the resulting higher incomes then stimulate consumption and hence further increases in aggregate demand (but by a lesser amount than in the preceding round), and so on, until the original impulse has eventually produced a multiplied effect on output and employment.

Within such a framework, fiscal policy appeared to be a powerful tool for regulating the level of employment, since a government deficit was thought to give rise to a multiplied increase in production and employment. Keynesians tended to believe that unemployment was a normal feature of an unregulated capitalist economy; but with the judicious use of fiscal deficits, the government could pump up the level of employment and achieve something resembling full employment. This became a fundamental premise of post-war social policy (Artis 1992, p. 139).

Later modifications somewhat softened the analysis of the powers of budget deficits, but did not reverse the basic thrust of the argument. It was recognized that a government deficit might raise interest rates, and insofar as these inhibited investment demand, this might offset some of the original expansionary impact of the deficit. The idea also grew that a reduction of unemployment owing to an expansion of aggregate demand might also lead to higher money wages and hence induce inflation. The notion of a Phillips curve trade-off between inflation and money wages (Phillips 1958), which was rapidly recast as a trade-off between inflation and unemployment, became a standard part of the arsenal. Fleming (1962) and Mundell (1963) extended the analysis to the relation between output and employment and the balance of trade ('external balance'). The resulting complexity of the analysis, with its multiplicity of potential 'targets' (desired levels of employment, inflation, interest rates, foreign trade balances, and so on), implied that economic policy was a task for the sophisticated. But it was clear that these complications were extensions of the basic theory, not a challenge to it.

Central to all of these developments was the notion that inflation would arise only when the economy was in the *vicinity of full employment*. But this confidently held conception had begun to break down by the late 1960s. By then, inflation had not only become a practical problem of some importance, it had also become a serious *theoretical* problem: whereas the Phillips curve predicted that

inflation would be accompanied by a fall in unemployment (which would stimulate a rise in money wages and hence prices), this actual new round of inflation was attended by a *rise* in unemployment. This new pattern seemed to contradict the very notion of a trade-off between the two.

One attempt to get around the difficulty was to suppose that expectations played a significant role in the wage–price spiral. Thus was born the concept of an expectation-augmented Phillips curve (Phelps 1967; Friedman 1968), along with a particular 'natural rate' of unemployment (NAIRU) which would keep inflation in check. Conflict models of inflation as well as the infamous NAIRU have their root in this same ground (Godley and Cripps 1983; Rowthorn 1984).

But these ideas proved to be of ambiguous benefit to the Keynesian paradigm: not only did they undermine the basic thrust of Keynesian social policy, they also provided the basis for the New Classical (NC) macroeconomics which was to eventually supplant Keynesianism itself (Artis 1992, pp. 140–2). For instance, the idea of a natural rate of unemployment has its roots in the automatic full employment paradigm of neoclassical economics – the very thing that Keynesians were trying to overthrow. In the neoclassical vision, it is assumed that when all markets are in equilibrium, all workers will be able to achieve their desired level of employment at some labour market-clearing wage. But if information is not quite perfect, and if there are impediments in the labour market, there will always be some frictional level, some 'natural' level, of unemployment even in general equilibrium (Mathews 1992, p. 247). Such a natural rate is *voluntary*, since it arises out the decisions of individuals not to work in the face of existing costs of job search and existing unemployment benefits, welfare benefits, and so on. Thus, contrary to Keynesian views, the mere existence of unemployment, even rising unemployment, did not prove that it was involuntary. Not surprisingly, Neoliberal economists have been quick to proclaim that existing unemployment was in fact all voluntary (Bennett 1995).

Secondly, it was claimed that the actual level of inflation depended not only on the level of unemployment but also on inflation expectations. A higher level of inflation expectations could give rise to a higher level of actual inflation at any given level of unemployment. Since inflation expectations were assumed to change slowly (exhibit persistence), it followed that it might be necessary to tolerate (perhaps even induce) an unemployment rate higher than the 'natural' rate for long enough to lower inflation expectations.

As these fell, they would lower the actual rate of inflation consistent with any level of 'unnatural unemployment' (unemployment in excess of the natural rate), thus permitting excess unemployment to be also reduced somewhat – until finally the economy would glide into a state of long-run equilibrium in which actual inflation and expected inflation would be zero, and unemployment would be at the natural rate (the lowest sustainable rate).

Keynesian economics might have accommodated the idea that 'wringing out' inflation could be costly. But the notion that any observed rate of unemployment was essentially voluntary was far afield from the original Keynesian conceptions. In any case, it was reality which once again dealt the decisive blow to Keynesian economics. Country after country in the capitalist world floundered in the 1970s and 1980s, exhibiting inflation, unemployment, slow growth and a rise in poverty and social misery – in spite of record budget deficits. These dismal patterns fuelled the growing sense that the Keynesian theory of fiscal policy, however much it had been modified, was simply inadequate in this new epoch.

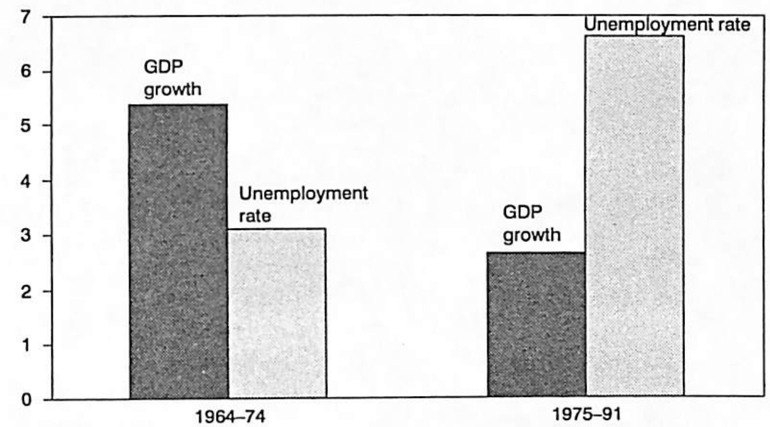
CONVENTIONAL THEORY VS. THE EMPIRICAL PATTERNS OF INFLATION AND UNEMPLOYMENT

We have seen that neoclassical and Keynesian theories differ on their explanation of inflation and unemployment. But it is important to note that they nonetheless share one a critical notion: namely, that there is an *empirical trade-off between inflation and unemployment.*

But is such a claim empirically supportable? Three points can be made here. First, as shown in Figure 4.1, between the first and second halves of the post-war period, the historical rise in average unemployment levels in OECD countries is directly associated with a corresponding fall in average output growth rates. I have tried elsewhere to show that this can be explained by the fact that a fall in the rate of profit gradually undermines the foundation for growth and hence produces the jump in unemployment rates (Shaikh 1987).

Second, as Figure 4.2 shows, there is *no general historical trade-off between unemployment and inflation*. As one can see, the patterns for the OECD countries as a whole indicate that while such a trade-off appeared to exist for the more recent period from 1975–91, the very opposite pattern holds for the early period from 1964–74 (comprehensive data is not available for unemployment before 1964).

Figure 4.1 OECD growth and unemployment, 1964–91



Source: *OECD Main Economic Indicators.*

Figure 4.2 OECD inflation vs. unemployment: 1965–91

Source: *OECD Main Economic Indicators.*

Indeed, this earlier pattern seems to have reasserted itself in recent times, as unemployment has fallen in countries like the United States with no discernible resurgence of inflation – much to the dismay of proponents of the natural rate hypothesis. By 1995 for instance, the US unemployment rate had fallen to 5.4 per cent, at a time when leading proponents such as Martin Feldstein and Robert Gordon had pegged the natural rate of unemployment – the trigger point of inflationary pressure – to be 6.0 per cent or even 6.5 per cent. Yet by 1997, with the unemployment rate still lower, there is still no evidence of renewed inflation (accelerating or otherwise). Gordon, at least, has responded by successively lowering his estimate of the natural rate as the actual rate fell below it (Bennett 1995).

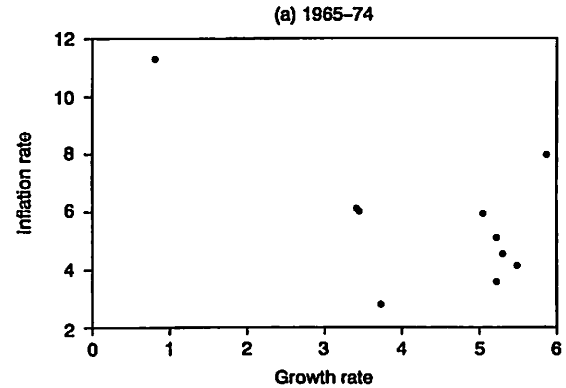
There is, however, an interesting clue in the empirical relation between inflation and economic growth depicted in Figure 4.3. In the first period from 1965 to 1974 (Figure 4.3a), even if one excludes the 1974 OPEC oil price jump in the upper left quadrant, there appears to be little relation between inflation and growth. If anything, it would suggest that lower growth is associated with *lower* inflation. But in the succeeding period from 1975 to 1991 (Figure 4.3b), lower growth is associated with *higher* inflation. As in the previous case, this behaviour is puzzling from the point of view of conventional theories. We will see that it need not be so in a classical–marxian theory of inflation.

RECONCILING THE EMPIRICAL EVIDENCE: AN ALTERNATIVE APPROACH TO INFLATION

The facts presented above are quite consistent with an alternative approach to inflation and unemployment derived from the classical and marxian traditions. There are three elements to this approach.

The first has to do with the question of short-run equilibrium. Both Keynesian and neoclassical economics tend to analyze actual output and the price level as if they were equilibrium levels associated with the short-run equality of demand and supply. From this point of view, the business cycle is a fluctuation in the short-run equilibrium output itself (Kalecki 1968). But I have argued consistently that the *process* of equalization of aggregate demand and supply is what gives rise to the observed three–five year business (growth) cycle: what we nowadays call 'the' business cycle is the fluctation in actual (disequilibrium) output as demand and supply

Figure 4.3 OECD inflation vs. growth



Source: *OECD Main Economic Indicators.*

Source: *OECD Main Economic Indicators.*

chase each other around an endogeneously generated growth path. This means that up and down phases of the cycle are associated with phases of positive and negative excess demand, respectively (Shaikh 1989, 1991, 1992).

The second element has to do with money and credit. Deficit spending by any unit – that is, its spending in excess of its current income – can be financed only by running down its assets (borrowing from stocks) and by borrowing from others (Earley, Parsons *et al.* 1976). For the economy as a whole, this boils down to the creation of new loans by private banks and new high-powered money by the central bank. Since neither new credit nor new high-powered money is created to satisfy the demand for money as a liquid asset, this can easily give rise to persistent episodes of aggregate excess demand fuelled by an endogenously generated excess supply of money (Moore 1989, p. 483). Aggregate deficit spending by governments, by the private sector (including households), combined with foreign inflows of purchasing power, can therefore result in persistent pressure on various markets, particularly the commodity market. In an ongoing project, we are developing measures of aggregate excess demand and of the finance behind it, and attempting to demonstrate their relation to growth and inflation in the US economy.

The third element involves the implication of persistent excess demand. Excess demand, which is the excess of a generally growing demand over a growing supply, accelerates the growth in supply. The limits to this process then arise from the limits to the growth in supply. Both neoclassical and Keynesian traditions assume that the availability of *labour* provides the general limit to commodity supply. Therefore they both anticipate that excess demand will stimulate inflation only after practical full employment has been attained. They differ, of course, on what practical 'full employment' means, and whether or not it is the normal state of capitalism, but they share the notion of a trade-off between unemployment and inflation. The trouble with this, as we have seen, is that it requires considerable contortions on their part to explain persistent periods of both rising inflation and rising unemployment.

Neither marxian theory nor capitalist history gives us any reason to suppose that output is limited by the supply of labour. Indeed, within the classical tradition there is a separate *intrinsic* limit to growth. In effect, *even when labour is freely available at the going real wage*, the maximum sustainable rate of capital accumulation within an economy is given by the normal-capacity rate of profit.

Marx was the first to demonstrate that sustained accumulation requires balanced growth, and it is clear from his schemes of expanded reproduction that the maximum sustainable growth rate occurs when all surplus value is reinvested – that is when the rate of growth equals the rate of profit (Marx 1981). One can arrive at a similar result along a Harrodian warranted (that is normal-capacity utilization) path with a (Kaldorian) classical savings function, since there the investment-savings equality $I = S = sc \cdot P$ implies that the warranted rate of accumulation of capital

$$gk^{w} = I/K = sc \cdot (P/K) = sc \cdot r$$

where sc = the savings propensity of capitalists, P = aggregate (normal capacity) profits, K = the stock of capital, and $r = P/K$ = the normal-capacity rate of profit.

It follows from this that the maximum warranted growth rate occurs when all profits are saved ($sc = 1$). Finally, the celebrated articles by von Neumann and Leontief demonstrate the existence of this same limit in multisectoral models (von Neumann 1945–6; Leontief 1953).[1]

I will call the maximum sustainable growth rate the 'throughput limit' of the economy. Now, suppose that in some period there exists persistent excess demand, along with unemployed labour. Then the excess demand will stimulate (accelerate) the rate of growth of output and of capital and reduce the unemployment rate – as long as the growth rate is not constrained by the throughput limit. *But if for any reason the gap between the actual growth rate and the throughput limit narrows*, there will less and less room for output growth and consequently more and more pressure on prices. The ratio of the actual accumulation rate to the throughput limit (the normal-capacity rate of profit r), which I will call the 'throughput coefficient', is therefore an index of inflationary pressure. Note that the throughput coefficient is simply the ratio of investment to normal-capacity profits, since the capital stock appears in the denominator of both the rate of accumulation (I/K) and the rate of profit (P/K).

The process described above need not come about through a rising growth rate. If the normal-capacity rate of profit were falling, as it has done for most of the post-war period in the United States, then one would expect growth rates of capital (which depend on expected profitability of investment), also to fall. But if the accumulation rate fell more slowly than the profit rate, the throughput coefficient (which is the ratio of the former to the latter) would *rise*. In this way, it becomes possible to understand how *falling profitability can induce both rising unemployment through slowed growth,* and also *increased inflationary pressure via a rise in the throughput coefficient*. I would argue that this precisely why most advanced economies experienced both stagnation and inflation in the 1970s and 1980s – something neoclassicals and Keynesians have had great difficulty in explaining.

To test the relationship between the throughput coefficient and inflation, one needs data on aggregate profits, capital stocks and capacity utilization. In what follows I will use data for the United States alone, because I have consistent series on the necessary variables.[2] It should be noted that the United States is a large part of the OECD as a whole. Figure 4.4 shows that both the US normal-capacity corporate profit rate and the corresponding rate of accumulation (growth rate of capital) fell sharply from the mid-1960s to the early 1980s. Such a fall explains the rise in unemployment rates over this period.

It is in Figure 4.5, which compares the US inflation rate with its throughput coefficient, that we find that the same movements also explain inflation over the period. The key empirical expectation is that the rate of inflation will tend to rise as the economy's accumulation rate approaches its throughput limit – that is, as the throughput coefficient rises. We can put this proposition to a crude empirical test by directly comparing the two. Figure 4.5 depicts the US rate of inflation (in terms of the GDP deflator), and the throughput coefficient defined here as corporate investment in plant and structures relative to total normal-capacity corporate profits. Normal-capacity profits are defined in a similar manner to normal-capacity (potential) output, by dividing actual profits by the level of capacity utilization, the latter being based upon a measure developed in Shaikh (1987).

It is quite striking how closely the inflation rate in the United States mirrors the movements of the throughput coefficient. Looking at Figures 4.4 and 4.5, we see that from 1947 to 1962 the profit rate is high and both the profit rate and the accumulation rate are stable. Therefore in this period the throughput coefficient is low and stable, and so is the inflation rate (and the unemployment rate). Then follows a brief Vietnam War induced profit boom from 1963 to 1965, in which the profit rate rises but the accumulation rate rises even more, so that the throughput coefficient rises substantially

Figure 4.4 US profit rate and accumulation rate (corporate sector, real rates), 1947–95

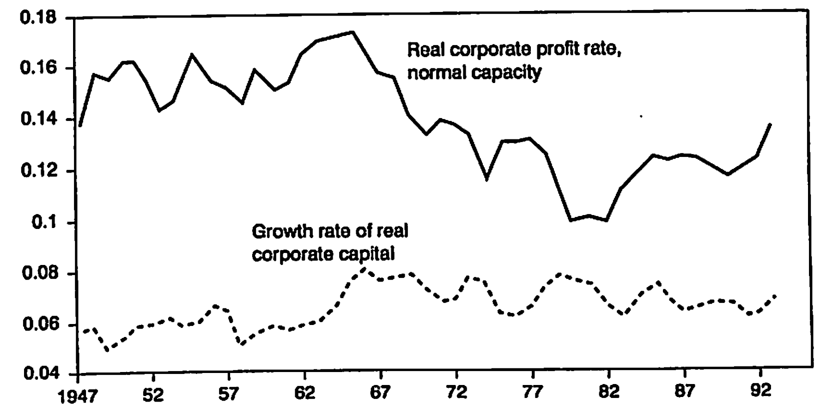
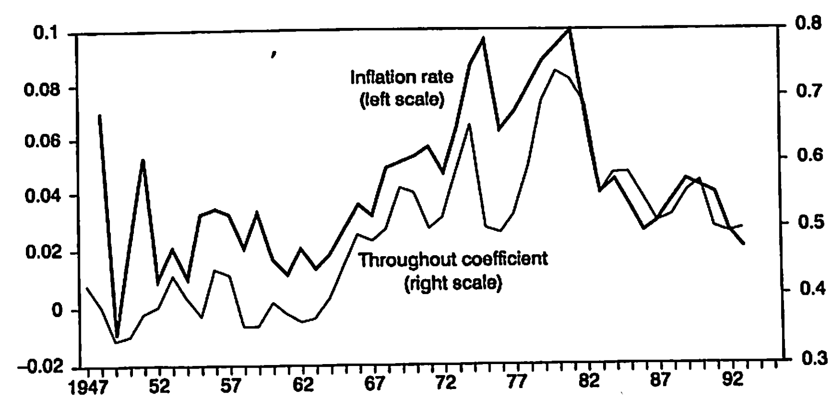


Figure 4.5 US inflation rate vs. throughout coefficient, 1947–95



– taking the inflation rate with it. From 1966 to 1982, however, the normal-capacity profit rate declines, and the the rate of accumulation follows suit. But the latter declines less rapidly, so that the throughput coefficient continues to rise, and hence so does the inflation rate. It is only in the last period, from 1983 to 1995, that a rising corporate profit rate manages to outstrip the accumulation rate, thereby sharply reducing the throughput coefficient. And it is precisely in this period that we find that the inflation rate falls just as sharply. On the whole, the throughput coefficient performs extremely well as an indicator of inflationary pressure in the US economy.

SUMMARY AND CONCLUSIONS

Both Keynesian and neoclassical theories expect that inflation will arise only in the vicinity of full employment. They differ among themselves on whether capitalism is normally in a state of full employment. But they share the critical notion that the expansion of supply is limited by the availability of labour, so that the pressure on prices increases as the system approaches full employment. This presumed trade-off between inflation and unemployment has been a central concern of both theory and policy over the post-war period.

But within marxian economics, no such presumption need exist. The concept of an endogenously generated and maintained pool of (involuntarily) unemployed workers is central to this tradition. This implies that the growth in labour supply will not generally provide the limit to the growth of output. And historical evidence certainly bears out the notion that inflation is not necessarily, or even usually, associated with (effective) full employment.

So how does one explain the fact that rising inflation was associated with rising unemployment in the 1970s–1980s, and that falling inflation is associated with unemployment remaining unchanged (in many OECD countries) or even falling (in the United States), in more recent times? I argue that the relevant limit to the growth of the system lies in its normal-capacity rate of profit, because that constitutes the maximum rate of accumulation (growth rate of capital) of the system. The ratio of the actual growth rate of accumulation to the normal profit rate, which I call the throughput coefficient, can therefore be viewed as strain guage for inflationary pressure.

The differential dynamics of the two variables involved provide the key to explaining inflation and its various links to unemployment. The data for the US economy over the postwar period bear this out by showing a strong connection between the throughput coefficient and the inflation rate (Figure 4.5).

Lastly, it is worth mentioning that although labour appears as the basic limit to production in a static framework, as in most accounts of Keynesian and Kaleckian theory, the notion of an intrinsic growth limit is perfectly compatible with dynamic versions of the very same theories. Harrod obviously comes to mind here. There need be no contradiction, therefore, between the ideas set forth here and those of the Keynesian and Kaleckian traditions. Indeed, the throughput coefficient, which is a kind of utilization rate of growth potential, frees us from the contortions involved in trying construct some mechanical trade-off between inflation and unemployment.

Notes

1. The aggregate Harrodian-type relation makes it clear that the same limit would exist even if over time the rate of profit were to change due to technical change and class struggle. Pasinetti, however, argues that in a disaggregated multisectoral model with ongoing technical change and changing demand proportions, the technology-based maximum rate of balanced growth derived by Leontief and von Neumann (1953; 1945–6) is no longer relevant (Pasinetti 1981, pp. 118–23). But while this may be strictly true, in the sense that ongoing differential rates of technical change and demand growth might modify the exact definition of the maximum sustainable growth rate (the throughput rate), it seems equally clear that they cannot abolish the limit itself.
2. Data for Figures 4.4 and 4.5 are from Citibase. Real investment = residential + non-residential investment, in 1987$. Real profits = total domestic corporate profits with IVA and Capital Consumption Adjustment, deflated by the implicit price deflator for investment. Normal capacity profits = real profits divided by capacity utilization. The measure for capacity utilization is derived from Federal Reserve Board survey data on capacity additions and expansion, as explained in Shaikh (1987, Appendix B). It was updated by regressing it on the published Federal Reserve Board series for manufacturing capacity utilization.

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EMPIRICAL ECONOMICS

Inflation and growth: Explaining a negative effect

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Abstract. The paper presents a monetary model of endogenous growth and specifies an econometric model consistent with it. The economic model suggests a negative inflation-growth effect, and one that is stronger at lower levels of inflation. Empirical evaluation of the model is based on a large panel of OECD and APEC member countries over the years 1961–1997. The hypothesized negative inflation effect is found comprehensively for the OECD countries to be significant and, as in the theory, to increase marginally as the inflation rate falls. For APEC countries, the results from using instrumental variables also show significant evidence of a similar behavior. The nature of the inflation-growth profile and differences in this between the regions are interpreted with the credit production technology of the model in a way not possible with a standard cash-only economy.

Key words: Endogenous growth, panel data, inflation, non-linearity

JEL classification: O42, C23, C51, E13

1. Introduction

Kormendi and McGuire (1985) document a negative effect of inflation on economic growth for a cross-section of 47 countries during the period 1950–1977. Recent panel evidence such as Barro's (2001) strengthens the support

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for such a negative effect. In qualification, Khan and Senhadji (2001), Ghosh and Phillips (1998), and Judson and Orphanides (1996) all find a significant negative inflation-growth effect above a certain "threshold" value of the inflation rate, and no significant effect below the threshold value, without using instrumental variables and with differences found between less and more developed country samples. Further the above-threshold negative effect that they find is significantly non-linear whereby the marginal effect is stronger at lower inflation rates than at higher ones; see also Fischer (1993).

Linking such evidence with a theoretical model has been somewhat tenuous. The Tobin (1965) paper concerns the induced increase in the capital stock within a Solow exogenous growth model. Stockman (1981) demonstrates a reverse Tobin effect of inflation on capital by applying Lucas's (1980) Clower constraint to investment as well as consumption purchases, but does not explicitly deal with the balanced-growth rate of output.[1] Sidrauski's (1967) money-in-the-utility function model and Ireland's (1994) AK model derive only a transitional effect of inflation on the growth rate. Dotsey and Sarte (2000) use an AK–type model, apply the Clower constraint to investment as in Stockman (1981), add uncertainty, and calibrate a negligibly small negative effect of inflation on the balanced-growth rate. Chari, Jones and Manuelli (1996) calibrate, in an endogenous growth model with human capital, a negative effect of inflation on the balanced-growth rate, but one of a magnitude far below what they consider to be within the estimated range found in empirical studies.

However, it has sometimes been overlooked that Gomme's (1993) endogenous growth framework, with a somewhat extended model relative to Chari et al. (1996), uses a Lucas (1988) type production function for human capital investment and produces a significant negative calibrated effect of inflation on growth. Because this model produces effects that are within the range of the significant negative effects found in recent panel estimates (see Gillman and Kejak 2002), it is a plausible candidate for a theory of the inflation-growth effect.

The contribution of the paper here is that we extend a Gomme (1993)-type model, and apply the extension to guide and interpret the estimation of the inflation-growth effect in a comprehensive way. This produces a simple way to understand the effect of inflation on growth, and a linked way to estimate this effect that is not inconsistent with previous work. However, the empirical results differ somewhat in that the effect of inflation is found to be significant and negative at all positive levels of the inflation rate, when using instrumental variables for both developed and less developed country samples. Together with the model this offers a novel combination of a fully supported theory of the inflation-growth effect.

The extension made to the Gomme (1993)–type model concerns the exchange technology. Instead of a cash-only economy, here as in Gillman and Kejak (2002) both money and credit can be used to buy the consumption good. The credit production function is specified explicitly in an additional layer of "micro-foundations". This allows interpretation of different results

[1] However the Stockman (1981) model can produce a negative effect on inflation on the balanced-growth path for the special case of an "AK" production function; see also Haslag (1998) who uses a Stockman-related approach with growth effects.

between OECD and APEC samples using differences in the credit technology, or in the "financial development".

The model-based explanation is that inflation lowers the rate of return on capital, both human and physical. Ever since the Ramsey-Cass-Koopmans theory endogenized the savings rate of the Solow model by using utility optimization, the growth rate has depended on one variable: the rate of return to capital. Taxes that decrease that rate of return decrease the growth rate. Models that explain growth endogenously, with a Lucas (1988) human capital accumulation, further develop the theory by implying that the growth rate depends on the rate of return to human capital as well as physical capital, whereby the rate of return on all forms of capital must be equal in the balanced-growth equilibrium.[2] A tax on either form of capital induces a lower return in equilibrium on all forms of capital. When such endogenous growth models are set within a monetary exchange framework, such as those of Lucas (1980), Lucas and Stokey (1987), or McCallum and Goodfriend (1987), the inflation tax also will affect the rate of return on capital. In particular, as Chari et al. (1996) discuss, the inflation tax induces goods to leisure substitution that lowers the return to human capital, and the growth rate.

with endog s, growth = f(r)

With a parsimonious empirical theory, the paper here explains growth through factors that reflect the return to physical and human capital in terms of easily measurable variables. The real interest rate, a measure of the return to physical capital, is proxied with the investment rate in a way suggested by the theory. Any further changes across countries to this real rate, for example as caused by differing tax regimes, are accounted for via use of fixed country specific effects within the econometric model. This is important in that the model implies (see King and Rebelo 1990), that a tax on capital income directly reduces the growth rate; a tax on labor income can also affect the growth rate for example if there is any kind of tax or subsidy to the human capital investment sector. The inflation rate enters the econometric specification as the one systematic, easily measured, tax on human capital within each country that is suggested by the theory.

Another aspect of the empirical work here is that time effects are conditioned upon, as fixed parameters in the econometric model, and interpreted as being related to unexpected international changes in the inflation rate.

The model also includes a variable designed to capture transitional dynamics. There are several potential sources of transitional dynamics in the model. For one, note that in the Lucas (1988) human capital model, without the assumption of a human capital externality, the human capital index acts to explain endogenously the original Solow exogenous technological change, or "total factor productivity" shift parameter. Further, Solow-like transitional dynamics exist when the model includes physical capital in the production of human capital, as in King and Rebelo (1990) and as in this paper. Thus there are the Solow-like transitional dynamics whereby a below-balanced-path-equilibrium amount of capital, now human as well as physical,

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[2] In contrast the balanced-growth path return on capital is fixed at A in AK models, and inflation effects this return and the growth rate only through devices that somehow put capital under the cash-in-advance constraint, such as in Stockman (1981), Haslag (1998) and Dotsey and Sarte (2000); but then these have non-empirically consistent reverse-Tobin type effects whereby inflation causes substitution away from capital because inflation lowers its return directly.

leads to increases in capital and income levels on the transition towards the stationary state. Applying this representative agent model to a panel suggests that countries within an interlinked group, or "growth club", will converge to the same balanced-growth path, as in Barro (1997) for example. Therefore including the ratio of income in each country relative to that of the balanced-path "leader", for example the US, can capture the effect of the transitional dynamics on the growth rate.

Second, Einarsson and Marquis (1999) present transitional dynamics for a Gomme (1993)–type model and indicate how the income variable moves transitionally with changes in the inflation rate, as a result of changes in the exogenous rate of money supply growth. This suggests that should the leader of a growth club have an influence on the inflation rates of the other countries in the club, the other countries in the club would have transitional effects on income that would be relative to the leader to some extent. For these two sources of transitional dynamics, and assuming the US as the lead country, we follow the older exogenous Solow-growth literature (Kormendi and Meguire 1985), and the new endogenous growth literature (Barro 2001), and include as a variable the ratio of the level of income in the US to the level of income in each other country.

Instrumental variables (IV's), to test for possible endogeneity of the inflation and growth rates, are little used in the literature cited above. We test for such endogeneity and report results for all samples using an instrument suggested directly by the theoretical model. The paper follows the model's implication that the money supply is exogenous and largely determines the inflation rate along the balanced growth rate. Further, there is evidence that finds that the money supply growth rate Granger-causes the inflation rate, while the inflation rate in turn Granger causes the output growth rate, such as in Nakov and Gillman (2002). This suggests the use of money as an instrument for the inflation rate in a way consistent with the model.

Also, note that the theoretical model predicts a non-linearity in the inflation-growth effect, whereby the effect is marginally stronger at lower inflation rates than at higher ones, but negative everywhere. So at the Friedman optimum of a zero nominal interest rate the effect is marginally the strongest, and monotonically weaker thereafter. This non-linearity is explored using alternative specifications, these being the natural logarithm, quadratic, and spline functions.

Finally, it is noted that the optimization model does not dismiss the Tobin (1965) effect, but rather re-states it in general equilibrium terms. Here, in contrast to Stockman (1981), there is a positive Tobin-type effect. The endogenous growth, cash-in-advance, setting implies that the inflation tax reduces the return on human capital, and that the return on physical capital must adjust downwards in equilibrium. This adjustment requires an increased investment and an increased capital/labor usage across all sectors. This input realignment slightly mitigates the degree to which the return on human capital and physical capital must fall as a result of an increase in the inflation rate. Thus the Tobin effect here is the more efficient use of inputs given the higher tax on labor relative to leisure that results from an inflation rate increase. It means a higher physical capital usage relative to effective labor, and a slightly smaller decline in the balanced-path growth rate. However, the effect of inflation on the balanced-growth rate is still negative, in contrast to the exogenous growth, monetary, model of Tobin, or related models reviewed by Ahmed and Rogers (2000).

2. Endogenous growth monetary framework

The representative agent works in a constant-returns-to-scale (CRS) goods sector that employs physical capital and effective labor. Effective labor is defined as raw labor factored by the human capital (quality indexed). The agent also devotes recourses to two additional, implicit price, sectors. These are the CRS human capital production that involves the investment of capital and effective labor, and a credit services sector that involves only effective labor in a diminishing returns technology. The agent faces four constraints on the maximization of utility over goods and leisure in terms of the flow of human capital; the flow of financial capital that is comprised of money and physical capital; the stock of financial capital; and the exchange technology. The technology of the credit services sector is built into the cash-in-advance constraint.

At time t, denote the real quantities of output and consumption goods by y_t and c_t, and the fraction of time spent in leisure, in credit services production, and in goods production by x_t, l_{Ft}, and l_{Gt}. The share of physical capital in goods production is given by s_{Gt}. The stocks of physical and human capital and their depreciation rates are given by k_t, h_t, δ_k, and δ_h respectively. Denote the input prices of capital and effective labor by r_t, the real interest rate, and w_t, the real wage. The positive shift parameters of the production functions of goods, credit services, and human capital are A_G, A_F, and A_H. Nominal variables are the price of goods P_t, the stock of nominal financial capital Q_t, the stock of money M_t, and the lump sum government transfer of cash V_t that is a constant fraction σ of the money stock. In addition denote by d_t the amount of real credit used in making purchases. Given parameters ρ, β, ε, and γ are in the (0,1) interval, and $\alpha > 0$, and $\theta > 0$.

2.1. The goods producer

Let the output of goods be produced by the function

$$y_t = A_G(s_{Gt}k_t)^{1-\beta}(l_{Gt}h_t)^{\beta}. \tag{1}$$

The firm's first-order conditions set the market's real interest rate and real wage equal to the marginal products:

$$r_t = (1-\beta)A_G[(s_{Gt}k_t)/(l_{Gt}h_t)]^{-\beta}, \tag{2}$$

$$w_t = \beta A_G[(s_{Gt}k_t)/(l_{Gt}h_t)]^{1-\beta}. \tag{3}$$

2.2. The consumer problem

The consumer's current period utility function is given by

$$u(c_t,x_t) = c_t^{1-\theta}x_t^{\alpha(1-\theta)}/(1-\theta). \tag{4}$$

2.2.1. Income and human capital constraints

The nominal financial capital constraint is

$$Q_t = M_t + P_t k_t. \tag{5}$$

The nominal income constraint derives from setting the change in financial capital to zero. This sets income of $r_t P_t s_{Gt} k_t + w_t P_t l_{Gt} h_t + V_t + \dot{P}_t k_t$ minus expenditure of $P_t c_t + \delta_K P_t k_t$ equal to zero:

$$\dot{Q}_t = r_t P_t s_{Gt} k_t + w_t P_t l_{Gt} h_t + V_t + \dot{P}_t k_t - \delta_K P_t k_t - P_t c_t = 0. \tag{6}$$

Human capital is CRS produced, with capital not used in goods production $(1 - s_{Gt})k_t$ and time not used in leisure, credit services production, or goods production $(1 - x_t - l_{Gt} - l_{Ft})$. The investment in human capital is given by

$$\dot{h} = A_H(1 - x_t - l_{Gt} - l_{Ft})^{\delta} h_t (1 - s_{Gt})^{1-\delta} k_t - \delta_h h_t. \tag{7}$$

2.2.2. Exchange technology

Money and credit are perfect substitutes in purchasing the consumption good. This can be expressed by equating the sum of real money balances and total real credit to the aggregate consumption:

$$(M_t/P_t) + d_t = c_t. \tag{8}$$

Define by $a_t \in (0, 1)$ the fraction of purchases made with cash, so that

$$[(M_t/P_t)/c_t] + (d_t/c_t) \equiv a_t + (d_t/c_t) = 1. \tag{9}$$

This makes the so-called cash-in-advance, Lucas's (1980) "Clower constraint", merely a part of the description of the perfect substitutability of money and credit:

$$M_t = a_t P_t c_t. \tag{10}$$

The money supply progresses through the government transfer, which is assumed to be made at a constant rate σ:

$$M_{t+1} = M_t + V_t = M_t(1 + \sigma). \tag{11}$$

From Eq. (9), it is clear that the share of purchases made by credit is given by $1 - a_t$. Or the total amount of credit used can be expressed as

$$d_t = (1 - a_t)c_t. \tag{12}$$

Consider specifying the production of this credit using an effective-labor only technology, with diminishing returns, whereby $\tilde{A}_F$ is a function that depends on the level of consumption c_t:

$$d_t = \tilde{A}_F[c_t](l_{Ft} h_t)^{\gamma}. \tag{13}$$

With the shift parameter containing this dependence upon c_t, the function $\tilde{A}_F$ is specified as $\tilde{A}_F = A_F c_t^{1-\gamma}$, and the credit production function is Cobb-Douglas in $l_{Ft} h_t$ and c_t:

$$d_t = A_F(l_{Ft} h_t)^{\gamma} c_t^{1-\gamma}, \tag{14}$$

which can be written using equation (12) as

$$(1 - a_t)c_t = A_F(l_{Ft} h_t)^{\gamma} c_t^{1-\gamma}. \tag{15}$$

The rationale of the introduction of c_t into the total productivity factor is that the credit supplier, which in a decentralized framework can be thought of as a hypothetical firm similar to American Express, must take total economic activity as a given. The credit supplier can only hope to increase its share of the total activity that is being exchanged with credit.[3]

The nature of the externality is chosen by restricting the consumption velocity of money to be stable on the balanced growth path, as evidence suggests allowing for shifts due to financial innovation. Rewriting the Eq. (15) as

$$1/a_t = 1/[1 - A_F(l_{Ft}h_t/c_t)^{\gamma}], \tag{16}$$

velocity $1/a_t$ is stationary because l_{Ft} and c_t/h_t are each stationary. Balanced-path changes in velocity, such as increases that result from deregulation and innovation in the Finance sector, are captured when A_F exogenously increases.[4]

Solving for a_t from Eq. (16), and substituting into Eq. (10),

$$M_t = [1 - A_F(l_{Ft}h_t/c_t)^{\gamma}]P_t c_t, \tag{17}$$

gives the constraint that enters the consumer maximization problem, which can be found presented in full in the Appendix in Gillman et al. (2001).

Note that Eq. (17) can be solved for l_{Ft}, the total time devoted by the consumer in the role as credit producer. This "banking time" solution of the constraint presents an exchange constraint that can be viewed as being equivalent to a special case of the shopping-time economy, as for example in Lucas (2000). Except here the time spent in exchange activity is only that time that enters into the credit production function.[5] The advantage of this over the shopping time models is that those models are typically, as in Lucas (2000), calibrated so as to yield a constant interest elasticity of money. Here the interest elasticity rises in magnitude as the inflation rate goes up, as consistent with evidence (see Mark and Sul 2002); the rising elasticity constitutes the central feature used to explain the non-linear nature of the inflation-growth effect.

2.3. *The effect of inflation on the balanced-growth path*

The representative agent maximizes the discounted stream of the period utility of Eq. (4), subject to Eqs. (5), (6), (7), and (17).[6] The marginal rate of substitution between goods and leisure is

[3] Gillman and Yerokhin (2003) detail how this model is equivalent to an interpretation of an Beckerian household production economy with a production of exchange using the intermediate goods of money and credit; the exchange is itself also and intermediate good that is then combined with the goods output to yield the Beckerian household consumption good.

[4] See Gillman and Otto (2002) for an empirical investigation of velocity as given by this model.

[5] See Gillman and Yerokhin (2003) for a proof of the shopping-time/banking-time equivalence and for further discussion.

[6] Details of the equilibrium conditions for this economy are found in Gillman and Kejak (2002); existence and uniqueness of the equilibrium is proved in the case of no physical capital in Gillman et al. (1999).

$$\alpha c/xh = w/(1 + aR + wl_F h/c), \tag{18}$$

where R is defined as the nominal interest rate. The marginal rate equals the shadow price of leisure w divided by the shadow price of goods, $1 + aR + wl_F h/c$. The goods shadow price includes a goods price of 1 and a cost of exchange that is the sum of the average cash cost aR, and the average credit cost $wl_F h/c$. This relation shows that an increase in the inflation rate, which increases R directly, goes in the direction of causing c/h to fall relative to leisure, x, by a first-order effect. There are second-order changes of lesser magnitude that go in the opposite direction. In particular, a falls and w rises as the inflation rate goes up, but calibrations in Gillman and Kejak (2002) show that the rise in R ends up being dominant for levels of the inflation rate below hyperinflation, as typically defined, and the substitution goes from goods (normalized by human capital) to leisure.[7]

The equilibrium balanced-path growth rate g is characterized by

$$g \equiv \dot{c}/c = \dot{k}/k = \dot{h}/h = [r - \rho]/\theta, \tag{19}$$

and by the equality of the return of physical capital in goods production to the return on effective labor in human capital production:

$$r = (1 - x)A_H \beta (s_H k/l_H h)^{1-\beta} - \delta_h. \tag{20}$$

Equations (19) and (20) suggest that an increase in leisure x may have a significant effect on decreasing r and the growth rate.[8] With Eq. (18), these equations show how inflation can cause a negative growth effect through the increase in leisure.

Calibrations in Gillman and Kejak (2002) show that this negative effect is very significant and robust. It occurs for a wide range of parameters around the baseline, which is set by using standard values from the literature. For the non-standard parameters, mainly $\gamma \in (0, 1)$ of the credit production sector, the full range of values was experimented with and all yield the negative inflation-growth effect. For example, with $\gamma = 0.2$ (the baseline value as based on evidence in Gillman and Otto, 2002), an increase in the inflation rate from 5 to 15% causes a decrease in the growth rate by 0.27%. The non-linear nature of the inflation-growth effect is illustrated by increasing the inflation rate further from 15 to 25%; this causes a smaller decrease in the growth rate of 0.22%. Such a decreasing magnitude continues to result with continued inflation rate increases.

The non-linearity implies that the inflation-growth effect falls towards zero at high inflation rates. Depending on the calibration, for standard parameters the inflation rate at which the growth rate decrease becomes zero is between 100 and 200%. This is effectively above any stationary (non-hyperinflation) rate of inflation likely to be experienced in any given country.

[7] See Gillman and Kejak (2000) for a human-capital only version of the model, which enables a closed-form solution, and details of how the inflation-growth effect turns positive only for rates of inflation above the level at which the magnitude of the interest elasticity equals one.

[8] Long run evidence presented in Ahmed and Rogers (2000) supports a decrease in the real interest rate as a result of an increase in the inflation rate.

2.3.1. Non-linearity of the inflation-growth effect

The intuition for the non-linearity derives from the exchange technology. When the inflation rate is at a low level, the consumer uses mainly money and just a little amount of credit. The theory implies that the interest elasticity of money demand is very low in absolute value, or "inelastic", at low inflation rates, and that it becomes increasingly more elastic (more negative) as the inflation rate rises and the agent substitutes towards credit. With an inelastic money demand, the agent substitutes from goods to leisure, and a bit from money to credit when the inflation rate goes up. As the interest elasticity increases with increases in the inflation rate, the agent still substitutes from goods to leisure but increasingly substitutes towards the use of credit away from money. The rising interest elasticity, and the emergence of increasing substitution towards credit as the primary substitution channel, means that the agent relies less on the goods to leisure channel. Therefore leisure increases at a decreasing rate, and the growth rate falls by increasingly smaller amounts. The bigger increases in credit and the smaller increases in leisure, as the inflation rate rises, explains why the inflation-growth effect is of smaller magnitude at higher inflation rates.

However, note that if no credit is available, then $a = 1$ in the model and the interest elasticity of money remains low in magnitude even as the inflation rate rises. This gives a more linear inflation-growth effect whereby the decrease in the growth rate from increased inflation remains large even at very high inflation rates. Below such a case is considered as an alternative for explaining the evidence

2.3.2. Tobin effect and the savings rate

The Tobin effect here is a general equilibrium one along the balanced growth path whereby an increase in the inflation rate causes an increase in the input price ratio, w/r, and in the capital to effective labor ratio in both goods and human capital production. Calibrations show that the inflation rate robustly causes a decrease in the return to capital, r, as the return on human capital is forced down, and an increase in the real wage w, as a result mainly of the consumer using more leisure. This induces substitution from effective labor to capital, and produces the model's Tobin-type increase in capital intensity, even while causing a decrease in the growth rate.

The savings rate is shown in this model, in Gillman et al. (2001), to depend on the input price ratio, w/r, on leisure, and on the nominal interest rate. The effect of an increase in the real interest rate r is to increase the savings rate. And in equilibrium the savings rate equals the investment rate. It is on this basis that we proxy the effect of the real interest rate on the growth rate through the use of the investment rate. This abstracts from other effects such as the real wage, and so makes the investment rate an imperfect proxy of the real interest rate.

2.3.3. Financial development

The credit production function depends primarily on the degree of diminishing returns parameter γ. Calibrations show that as γ decreases from its baseline value, the inflation-growth effect is less. With the interpretation that less developed countries have a greater degree of such diminishing returns, then the inflation-growth effect is pivoted up for developing versus developed countries, as in Fig. 1 (see Sect. 6).

3. The data

Three panels of countries are examined. The first consists of 29 OECD countries; the second panel consists of 18 APEC members (six of them jointly belonging to the OECD); and the third panel includes all 41 countries.

The data are from *EconData* and *World Bank World Tables*. The data set comprises annual measures on the following four variables: Per capita GDP, 1995 \$US million; average annual growth rate of real GDP, percent per annum; GDP deflator, percent per annum inflation rate; and the proportion of gross domestic investment in GDP, percent (these series have the following names: PCGDP, GDPGR, GDPDEF, INVPGDP respectively). The money stock data series is that listed as "money" in the online International Financial Statistics, or as M1 for some countries with multiple monetary aggregate listings; these generally correspond to the M1 aggregate.

The original sample period is 1961–1997 for all countries, except the Czech Republic (1985–1997), Germany (1992–1997), Turkey (1969–1997), Poland (1985–1997), Russia (1990–1997) and Vietnam (1986–1997).

Data is smoothed by setting each 5-year period as one averaged observation, as for example in Ghosh and Phillips (1998).

There appears to be no one definitive measure of the variable that represents the "inflation rate" in the literature. For example, Barro (1995) uses the inflation rate level, π; while Judson and Orphanides (1996) use the logarithm form, $\log(1+\pi)$. Ghosh and Phillips (1998) use four measures: π; $\pi/(1+\pi)$, $\log(1+\pi)$ and a non-monotonic transformation, $(1/(1-\gamma))\pi^{(1-\gamma)}$; Khan and Senhadji (2001) use $\log(\pi)$. In the raw data, inflation rates range from −11% to over 6,000%. To avoid hyperinflation measures, the data alternatively is capped at an inflation rate of 50, 100, and 150%, whereby values with higher rates are dropped from the sample. Also the few negative observations are dropped. Examination of the distributions shows that substantial outliers are still heavily skewing the distribution of the level of the inflation rate, such that this may bias the estimated inflation effect. The $\log(1+\pi)$ has a more normal distribution of inflation rates.

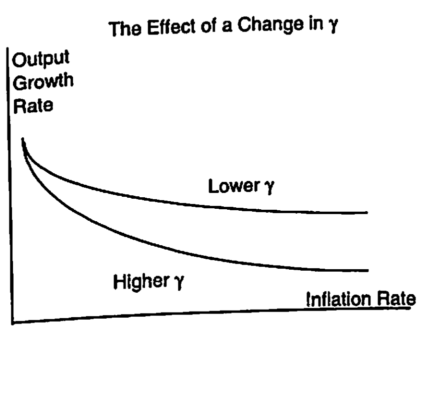

Fig. 1. Comparative statics of the credit production function

4. The econometric model

The economic model derived in Sect. 2, leads to the following econometric specification

$$y_{it} = \alpha_i + \lambda_t + \beta_g g(\pi_{it}) + \beta_{I/y} \ln(I_{it}/y_{it}) + \beta_{y/y} \ln(y_{USA,t}/y_{it}) + u_{it} \tag{21}$$

where: y_{it} is the average annual growth rate (% pa) in GDP at constant prices, of country i in year t; $\underline{\beta}$ the vector of unknown coefficients; $g(\pi_{it})$ a non-linear function of the annual rate of inflation; I_{it}/y_{it} the proportion of gross domestic investment in GDP (equal to the savings rate in the representative agent framework); $y_{USA,t}/y_{it}$ the ratio of US output to country i output; α_i the country specific, time invariant, effect which captures unobserved country heterogeneity, such as physical tax rates (conditioning on such, allows long-run growth rates to differ across countries, irrespective of their observed heterogeneity); λ_t the country invariant time effects, which account for any trend-deviation effects; and u_{it} the disturbance term. Signs on the investment/saving rate and on the ratio of incomes are predicted to be positive, while the inflation effect is predicted to be negative.

A fixed effects approach is followed in the estimations, as in Matyas and Sevestre (1996), to avoid any potential biases arising from correlations between the included variables and the unobserved effects.

4.1. Modeling the inflation-growth non-linearity

Several variants of the non-linear relationship between π and growth, $g(\pi_{it})$, were estimated. First, as in Judson and Orphanides (1996) and Ghosh and Phillips (1998), the relation is specified as $g(\pi_{it}) = \log(1 + \pi_{it})$. In addition, a "tied" spline is constructed. For the log specification, the spline is $g(\pi_{it}) = \sum_{j=1}^{3} D_j \beta_j \log(1 + \pi_{it})$, where D_j are three dummy variables, with D_1 representing "low", D_2 "medium," and D_3 "high" inflation. Restrictions are imposed on the parameters to ensure that the spline functions are continuous at the spline knots. In addition to these two specifications for the logarithmic function, there is an instrumental variables estimation using the money stock as the instrument.[9]

4.2. Robustness and endogeneity

Experiments were conducted regarding the robustness of the specification of the econometric model, the cut-off point for inflation rate outliers, and possible endogeneity bias from any simultaneity of growth and inflation. In terms of the robustness of the conditioning variables, the literature reports experiments with a variety of conditioning sets. That is, in addition to inflation, different sets of explanatory variables are included in the econometric speci-

[9] Extending the linear model, a quadratic form was also estimated whereby a squared term is included in π. Here $g(\pi_{it}) = \sum_{j=1}^{2} \beta_j \pi_{it}^j$ so that $g(\pi_{it})$ is a quadratic in the level of inflation. For this, the spline specification is $g(\pi_{it}) = \sum_{j=1}^{3} D_j \beta_j \pi_{it}$. The results did not improve upon the logarithmic specification and are not reported here but can be found in the working paper, Gillman et al. (2001).

fication (for example human capital variables). The different conditioning sets tend to be insignificant in terms of their effect on the inflation-growth relationship (see, for example, Khan and Senhadji 2001). While also experimented with here, such additional variables were not found to be significant and such results are not reported.

Negligible differences were found when using the different inflation rate truncation points of 50%, 100% and 150%. Given the traditional (Cagan 1956) hyperinflation definition of rates over 50%, the base specifications reported below use the 50% cut-off point.

The inflation rate enters the econometric model of Eq. (21) under the assumption that it is an exogenous variable relative to the output growth rate. To try to eliminate potential endogeneity bias, the model is re-estimated by the use of instrumental variables. Ghosh and Phillips (1998) use three alternatives: central bank independence, the exchange regime, and central bank governor turnover. They find a negative significant inflation-growth effect for the first two of these and insignificance for the latter. The approach here is to use current and lagged values of the money supply as instruments for inflation. This is suggested from the Sect. 2 model's balanced-growth equilibrium in which the money supply is exogenous and causes changes in the inflation rate.[10]

5. Results

The results reported in Table 1 are with data observations dropped from the sample if the inflation rates are above 50%. The table reports the results for the case when the inflation rate enters in the log form, $\ln(1+\pi_{it})$ (Specification A), for the IV version of the log specification (Specification B), and for the spline approximation of a non-linear relationship in the log function without IVs (Specification C). Figures 2–4 illustrate the results, with the "growth" label referring to Specification A, the "IVs" label to Specification B and the "Spline" label to Specification C.

Robust standard errors are reported in each case. In all specifications one rejects the null hypothesis that the individual and time effects are jointly zero. The direction and shape of the inflation effect in Table 1 is clear for Specifications A and B. A negative effect on the variable $\log(1+\pi)$ implies a non-linear negative relationship. For the spline, the results can be readily interpreted in terms of the implied inflation-growth profiles, in Figs. 2–4. Here the splines are represented according to their various implied marginal effects with all other variables evaluated at appropriate sample means.

For the OECD group of countries, there exists a striking amount of consensus of the non-linear negative inflation effect, irrespective of the estimation technique and the specification of the inflation effect in the estimated equation. Figure 2 shows that the marginal negative effect of inflation on growth is greatest at low levels of inflation – in particular at levels below

[10] As there were more missing values in these series, the IV versions generally have smaller sample size. Sargan-like tests for the appropriateness of such instruments are difficult to implement in fixed effects models, and are not reported for example in Ghosh and Phillips (1998) IV estimation (Table 5).

Table 1. Logarithm of inflation, logarithm of inflation – IVs and spline function in the logarithm of inflation

	OECD Coefficient	FULL Coefficient	APEC Coefficient
Specification A: *Within* Estimation; $g(\pi_{it}) = \ln(1+\pi_{it})$			
$\ln(I_{it}/y_{it})$	0.260 (0.026)*	0.220 (0.020)*	0.232 (0.031)*
$\ln(y_{it}^{USA}/y_{it})$	3.059 (1.654)**	2.196 (1.185)**	3.168 (1.589)*
$\ln(1+\pi_{it})$	−0.774 (0.132)*	−0.427 (0.123)*	−0.060 (0.218)
Constant	−1.717 (0.837)*	−1.180 (0.896)***	−2.668 (1.786)***
$\overline{R}^2$	47%	48%	43%
F-test	9.254*	8.598*	4.775*
NT	932	1,253	528
Hausman	3.813*	6.176*	5.162*
Specification B: IV *Within* Estimation; $g(\pi_{it}) = \ln(1+\pi_{it})$			
$\ln(I_{it}/y_{it})$	2.255 (0.487)*	2.765 (0.428)*	3.289 (0.607)*
$\ln(y_{it}^{USA}/y_{it})$	−5.190 (1.541)*	−3.939 (1.146)*	−2.010 (1.585)***
$\ln(1+\pi_{it})$	−0.922 (0.168)*	−0.617 (0.147)*	−0.448 (0.236)**
Constant	2.120 (0.944)*	1.720 (0.897)**	1.287 (1.435)
$\overline{R}^2$	44%	46%	46%
NT	835	1,086	458
Specification C: *Within* Estimation of the Spl ine Function; $g(\pi_{it}) = \sum_{j=1}^{3} D_j\beta_j \log(1+\pi_{it})$			
$\ln(I_{it}/y_{it})$	0.258 (0.026)*	0.213 (0.020)*	0.219 (0.031)*
$\ln(y_{it}^{USA}/y_{it})$	3.635 (1.674)*	2.532 (1.190)*	3.347 (1.590)*
π_{it}	−0.567 (0.164)*	−0.182 (0.155)	0.222 (0.272)
$(\ln[1+\pi_{it}] - \ln[10]) \times 1(\pi_{it} > 10)$	−1.053 (0.565)**	−1.117 (0.544)*	−0.912 (0.971)
$(\ln[1+\pi_{it}] - \ln[20]) \times 1(\pi_{it} > 20)$	0.589 (1.153)	0.297 (1.039)	−0.849 (1.767)
Constant	−2.037 (0.849)*	−1.431 (0.901)***	−2.827 (1.789)***
$\overline{R}^2$	47%	48%	43%
F-test	9.024*	8.588*	4.856*
NT	932	1,253	528

Notes: *p*-value of *F*-test for joint significance of all of the unobserved (fixed) effects (null model, are jointly zero); *Hausman* is the Hausman test for endogeneity of the inflation variable (null model is of exogeneity); robust standard errors in parentheses.
* Reject (two-sided) null hypothesis at 5% size.
** Reject (two-sided) null hypothesis at 10% size.
*** Reject (one-sided) null hypothesis at 10% size.

around 10%. Moreover, in each of the separate splines, the inflation effects are individually significant, at least at the 10% level, with one exception being the high inflation section of the spline function in the log specification. Instrumental variables estimation gives an almost identical result to those without instruments, suggesting little effect on the inflation rate coefficient of any endogeneity between inflation and growth.

When only APEC countries are considered, illustrated in Fig. 3, further reductions in significance levels are witnessed, and the expected non-linear relationship is only somewhat evident in the logarithm specification. The only

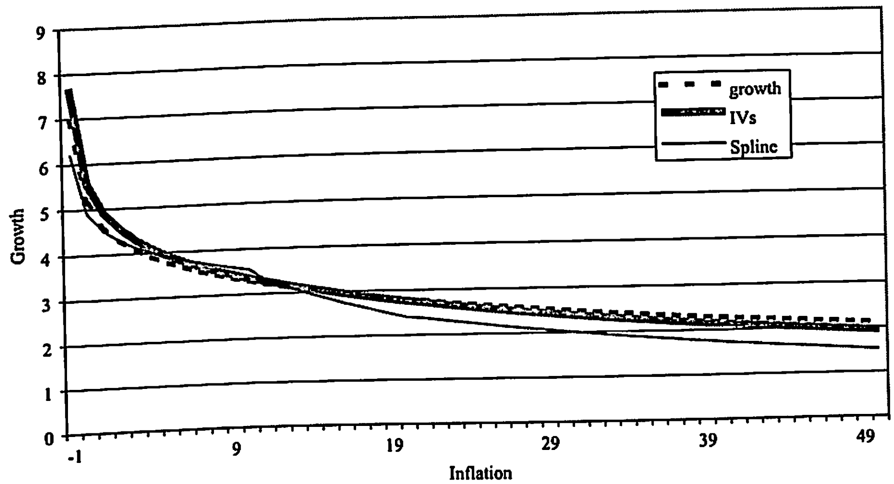

Fig. 2. Inflation-growth relationship, OCED, Inflation < 50%

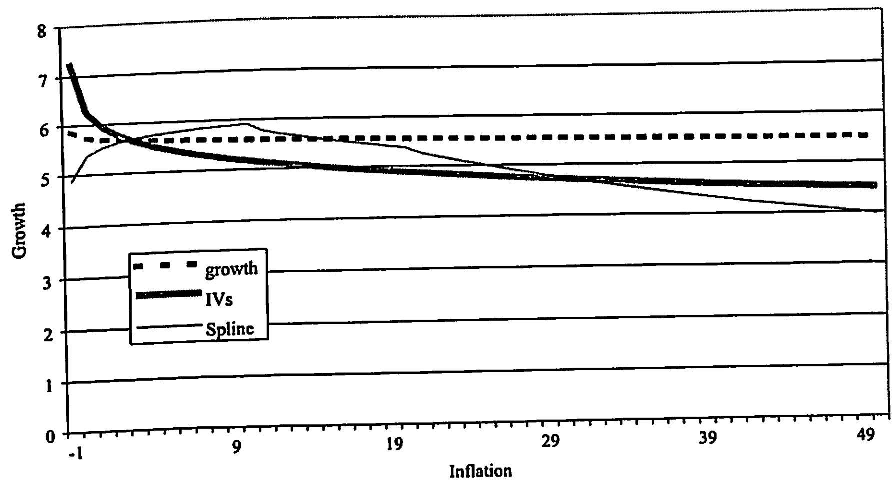

Fig. 3. Inflation-growth relationship, APEC, Inflation < 50%

significant inflation effect comes with the IVs estimation, with a negative inflation-growth effect at all levels of the inflation rate. Here the coefficient is about half of that of the OECD inflation-growth effect. For the splines estimation, in Fig. 3, there is a positive but insignificant effect at low levels of inflation. This insignificant effect is increasingly negative at levels of inflation in excess of 10%.

The full sample results in Table 1 show a lower significance level than the OECD results, for the inflation variables, in one case. This is for the spline results in the range of the lowest inflation rates. It reflects the APEC positive but insignificant results in this range. Also, the full sample coefficient of the inflation effect in the IVs estimation is about halfway between the level of the OECD and APEC coefficients. These results can be summarized as saying that the OECD results show up in the entire sample, but with less robustness.

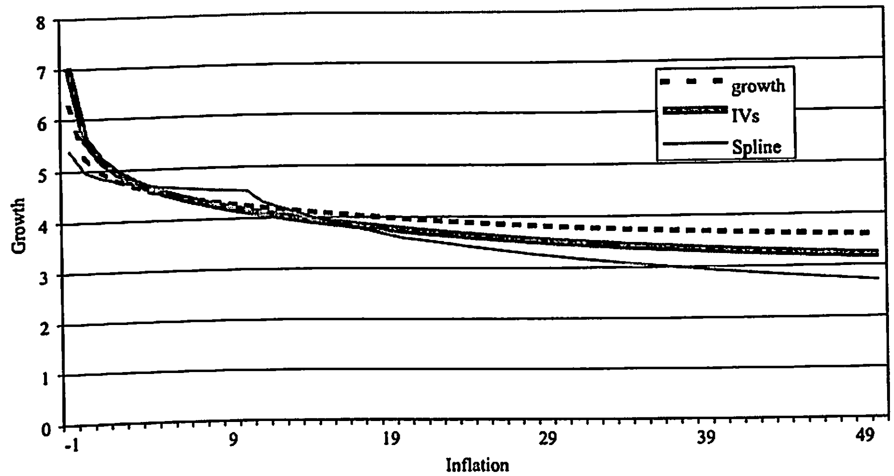

Fig. 4. Inflation-growth relationship, Full sample, Inflation < 50%

The non-linearity still emerges in all three specifications, as shown in Fig. 4, with it most pronounced in the logarithm specification.

Taken together, the results show the importance of separating out the OECD from the APEC countries, in the sense that the negative effect of the inflation rate is more robust and stronger in the separate OECD sample. However, instrumental variable estimation finds that the theoretically-predicted effect still is operative in the generally less financially-developed APEC group.

6. Discussion of results

6.1. Comparison

The results compare closely to those of Khan and Senhadji (2001), Judson and Orphanides (1996), and Ghosh and Phillips (1998). First, our use of the logarithm specification has support in Ghosh and Phillips (1998) who find this specification to have the best fit in non-linear models, while finding that the alternative linear specification is biased. Second, all of these studies above find significant negative inflation-growth results for rates above their chosen threshold values, for all samples. Third, for rates of inflation below the threshold values, using splines and no instrumental variables, all three report a positive but insignificant effect of inflation on growth for non-OECD (or lower income) samples, and a negative inflation-growth effect for OECD (or high income, or developed) country sub-samples.

Khan and Senhadji (2001) find an optimal threshold value inflation rate for developing nations sample of 11% with a positive and insignificant inflation-growth effect for rates below the threshold, without using instrumental variables. This corresponds to our spline results in Fig. 3, which show a somewhat positive but statistically insignificant effect of inflation for inflation rates 10% or below. Khan and Senhadji (2001) further find a significant negative effect for all inflation rates above one percent for developed countries, similar to both our non-instrumented and instrumented results in Fig. 2. Judson and Orphanides (1996) use a 10% threshold value below which they find positive and insignificant effects for their non-OECD sub-samples, similar to our Fig. 3 spline results. And for the OECD at inflation rates below the threshold, they find a negative insignificant inflation-growth value; this compares to our Figure 2 spline except that our results are significant.[11]

Ghosh and Phillips (1998) pick a two and a half percent inflation rate as the threshold value and find a positive inflation-growth effect for lower income sub-samples of countries, and a negative inflation-growth effect for an upper income sub-sample; again these non-instrumented results compare directly to our APEC and OECD log specification splines. When they use exchange rate and central bank independence instrumental variables, they report significant negative inflation-growth effects for their full sample (sub-samples not reported); this compares to our negative IV effects in all log specifications.

6.2. Characterization

To interpret the results, first consider their characterization. Consider a comparison of any of the three alternative OECD log specifications in Fig. 2 relative to the APEC IV log specification in Fig. 3. The initial slope of the APEC profile, at the lowest inflation rate, is not as steep as the comparable OECD slope. And while both the APEC and OECD slopes show significant non-linearity, it can be seen that the slope of the APEC profile is flatter at all inflation rates. Thus the APEC profile is less non-linear and similar to a "pivoting up" of the OECD profile.

6.3. Interpretation

There are two dimensions of the credit model that aid in the interpretation of these results. First is the general extension of the cash-only model. To see this, consider an explanation 1) using the cash-only economy which is the special case of $a = 1$, and 2) the extended model of Sect. 2 with credit being produced. Calibrations in Gillman and Kejak (2002) show that the model with cash-only has a bigger inflation-growth effect but a substantially more linear one, unlike any of the profiles in Fig. 2. When the inflation rate rises, in the cash-only model the magnitude of the interest elasticity rises slightly, while with the provision of credit in the model the magnitude of the interest elasticity of money rises much more quickly, making the inflation-growth effect increasingly much smaller, as in Fig. 2. A simulated cash-only economy inflation-growth profile, as based on a calibration such as in Gillman and Kejak (2002), if superimposed on those profiles of Fig. 2, would "cross" the other lines at moderately high levels of the inflation rate because the growth rate would fall too quickly relative to the evidence. In contrast the credit

[11] Khan and Senhadji (2001) note in their conclusions specifically that the inflation and output growth rates may be endogenous, so that non-instrumented results may entail biased estimate coefficients.

model simulation shows a profile that tapers out towards a zero inflation-growth effect, as in Fig. 2.

Second, the use of the credit production model has the additional advantage in that the nature of financial development within the economy can be described, and the interest elasticity determined, by the credit technology parameters. This allows differences in such parameters to be used to potentially explain differences in the inflation-growth profiles between the OECD and APEC results. Figure 1, showing two inflation-growth profiles with a different degree of the diminishing returns parameter, γ, corresponds well to the differences found between the OECD and APEC profiles with IV. Since $\gamma = 1$ is the case of constant returns to scale, a smaller γ indicates a greater degree of diminishing returns. It is plausible that the less developed economies would have a greater degree of diminishing returns, which would suggest assigning a lower value of γ to APEC than to the OECD. Calibrations in Gillman and Kejak (2002) show that a smaller γ implies a higher level of the interest elasticity of money. So as the inflation rate rises from low levels, the initial decrease in growth would not be as severe. And as inflation continued to rise, the higher interest elasticity would make the profile for APEC have a flatter slope at all levels of the inflation rate, as in the IV estimation of Fig. 3.

7. Conclusion

Our results confirm findings in the literature of a significant negative inflation-growth effect. Further our non-instrumented results replicate the insignificant positive inflation growth effect at low inflation rates for developing countries. Use of instrumental variables is not common in literature. Ghosh and Phillips (1998) report the use of IVs for one sample and find significant negative inflation-growth effects, but do not distinguish between regions. Our results suggest that the instrumental variables make a difference for the APEC but not the OECD sample. And these instrumented results show a negative inflation-growth effect at all levels of the inflation rate in all samples.

The results indicate interesting differences between regions that we interpret through the use of the endogenous growth human capital economy, with credit production explicitly modeled. With the advantage of the additional credit sector microfoundations, the paper presents a way to explain comprehensively the inflation-growth empirical evidence with a theoretical model. In particular, the inflation tax causes substitution to leisure at a decreasing rate as the inflation rate rises and the magnitude of the interest elasticity of money likewise rises. This causes a decreasing negative inflation-growth effect, at all levels of the inflation rate, and causes its highly non-linear inflation-growth profile. This is a human capital driven growth model, as in Lucas (1988) and as used extensively in the macroeconomic literature, but the paper's contribution is to use it to generate and interpret the inflation-growth profile.

The theory is consistent with a positive realignment towards a higher capital to effective labor ratio as a result of inflation, what we consider to be a generalized Tobin effect. This occurs even while the growth rate goes down. Such realignment has found empirical support in Gillman and Nakov (2003), for the postwar US and UK; see also Ahmed and Rogers (2000) for additional long term US evidence in support of a Tobin effect. The growth evidence of this paper, given its theoretical model, makes it a complement to the Tobin evidence.

The model also attributes a significant role to the leisure substitution away from employment as a result of inflation. While there is no unemployment per se in the model, the change in employment makes this relation similar in some ways to a reverse long-run Phillips curve. Further, there is evidence that is consistent with this. Ireland (1999) finds that inflation and unemployment are cointegrated, while testing a Barro-Gordon hypothesis that unemployment causes inflation. However he presents no evidence on causality and so these results equally support the model here in which inflation (really an increase in the money supply growth rate) causes a decrease in employment. Further, Shadman-Mehta (2001) finds similar inflation and unemployment cointegration, and also presents evidence that inflation Granger causes unemployment. This study uses long historical series for the UK including Phillips's original sample sub-period. Parallel to the growth and Tobin- type evidence, the employment aspect of the model could be further investigated for example by using the theory to give the structural VAR in Shadman-Mehta (2001) even more structure, and by also studying other countries ideally within a panel.

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What Caused Worldwide Inflation: Excess Liquidity, Excessive Credit, or Both?

By

James S. Earley

Contents: I. Introduction. — II. When Money Mainly Mattered. — III. Simple (Monetary) Financial Intermediation. — IV. The Dollar Currency System. — V. The Growth of Liability Management. — VI. The Eurodollar Markets. — VII. Eurodollar Funds from Non-Demand Deposit Sources. — VIII. Conclusions.

I. Introduction

This paper seeks a partial reconciliation of the "Monetarist" and the "Keynesian" variants of the "demand-pull" theory of the genesis of worldwide inflation[1]. It seeks a middle-ground, superior to both.

Monetarism sees the cause of the inflation to lie in an inordinate growth of *monetary* liquidity, causing the world's non-financial sectors to spend beyond the world's capacity to produce. Typical "Keynesianism" sees the major cause of excess effective demand not in excessive liquidity but in other, mainly non-monetary, causes. The position of this paper, very broadly stated, is that excessive, mainly *non*-monetary liquidity, *engendered by new forms of "financial intermediation,"* so increased worldwide credit availability and credit extension as to make world spending excessive.

My position is essentially a development of the now-classic Gurley-Shaw [1956] theory of the influence of financial intermediation. The paper stresses the developments in intermediation that have occurred since Gurley and Shaw set forth their theory, and especially such developments in international finance.

Remark: This paper is a revision of one presented at the annual meetings of the Western Economics Association on June 22, 1978. The author wishes to acknowledge with great thanks the help of the following in the evolution and publication of the paper: Joseph Coppock, Paul T. Ellsworth, H. Robert Heller, Randall Hinshaw, George McKenzie, Walter Salant, the Managing Editor of this Journal, and an anonymous referee. The help of each was substantial, but none of them is responsible for the paper's remaining shortcomings.

[1] The elements I stress lie on the "demand-pull" rather than the "cost-push" side of what I believe must now be a dual-based explanation of our ongoing inflation. Although the cost-push factors may by now have become even more powerful, the demand-pull forces, I believe, were primarily responsible for its genesis and development.

17 (21)

Viewed differently, my diagnosis represents an amendment, although a basic one, to the monetarist theory of world inflation as expressed by the late Harry Johnson [1972]. He wrote (pp. 55 f.) that the monetarist position "rest[s] on the fundamental assumption that a highly developed financial system permits a high degree of substitutability of one type of expenditure for another and one source of finance for another, *the ultimate constraint on expenditure being the stock of money on which the pyramidal structure of credit creation rests.*" (Italics supplied.)

Unlike some other monetarists, it will be noted, Johnson recognized the importance of financial structure, and also the role of credit. These are the matters I stress. Recent developments in international (and also domestic) finance have virtually removed the anti-inflationary constraint that the "stock of money" once exercised on credit extension and hence on world spending.

I have elsewhere set forth the basic theoretical framework underlying my thesis in the context of a closed economy, and tested it, provisionally, by reference to US Flow of Funds data [Earley *et al.*, 1976]. This paper applies similar theory to the international economy.

My theory sees changes in aggregate money spending in a closed economy to turn on "deficits" and "surpluses" run by micro-units or sectors of the economy. Sharply rising aggregate expenditures are seen as resulting from initial financial deficits, run by units or sectors, that are matched by the (axiomatically necessary) financial surpluses of other units and sectors only after there has been sharply increased total spending. The major source of financing these deficits, in a well-developed financial economy, is seen as borrowing. The main lenders are bank and non-bank financial intermediaries, whose sources of loan funds have now become only very loosely linked to either money stocks or current saving.

It is a very natural thing to transpose this approach to the international economy, treating national economies as the sectors among which sources and uses of funds transactions, which are the technical backbone of the analysis, take place. This is what I do in most of what follows.

My thesis is that the worldwide inflation of recent decades was caused by the combination of an unusually strong desire of governments, business firms and households to run deficits — i.e., to spend on goods and services in excess of their current receipts on goods and services account — plus an unprecedented facility of financing these deficits because of innovations in domestic and international financial intermediation. The three major financial innovations that permitted this growth of worldwide "credit availability" were: first, the Dollar Currency System; second, Liability

Management; and third, the Eurocurrency markets, especially the Eurodollar market[1].

II. When Money Mainly Mattered

I introduce my analysis with a simple sources and uses of funds table of international trade and finance in a two-nation world devoid of financial intermediation. The triple purpose of this initial exercise is: (1) to introduce the sources and uses of funds technique, which will be used throughout the paper; (2) to show how even non-intermediated credit tends to expand

Table 1 — *Sources and Uses of Funds Analysis: Balanced International Trade and International Deficits and Surpluses Settled by Money Transfers and Direct International Credit*

	Country A		Country B		World economy	
	uses	sources	uses	sources	uses	sources
Balanced trade	imports 100	exports 100	imports 100	exports 100	imports 200	exports 200
Increased international (deficit) spending by Country A settled by:						
(a) Transfer of money	+ imports 20	— money 20	+ money 20	+ exports 20	imports 220	exports 220
(b) International credit	+ imports 20	+ liab 20	+ loans 20	+ exports 20	imports 220	exports 220
					+ loans 20	+ liab 20

liab = liability

Notes: It is assumed that the loans take the form of direct (trade) credit from exporters of Country B to importers of Country A. The imports of Country B in the top panel (100), and the corresponding exports of Country A (100), are assumed to continue in the bottom panel. This panel does not of course show "equilibrium" positions in either (a) or (b). The imbalances would be expected to lead to changed spending in either or both countries.

[1] It should be borne in mind that the running of deficits, and the credit that financed them, occurred within as well as between national economies. To focus on international finance we neglect the lower and larger portion of the credit iceberg. But in recent times international credit is a great deal more than the iceberg's tip.

spending; and (3) (especially) to show why credit without intermediation is unlikely to lead to serious inflation.

In the top portion of Table 1 international trade is balanced. No net money or credit flows are involved. If the two nations use different units of account, foreign exchange transactions take care of the conversions. In the lower portion of the table Country A increases its international spending, running a financial deficit. Under (a) this deficit is shown as settled by the transfer of commodity money (e.g., gold). In (b) it is settled by the granting of direct trade credit from exporters in B to importers in A. In (a) the source of funds for the increased imports of Country A is the running down of its money balances. In (b) it is trade borrowing.

One salient difference between the two cases is that in (a) the deficit country loses cash liquidity, whereas in (b) it merely has a liability to pay a debt at some future time. Case (b) is therefore less likely than (a) to bring about a reactive contraction of international (and domestic) spending by Country A.

Surplus Country B, on its side, enjoys increased export earnings in either case, but in (b) it holds a debt rather than increased money stock. It is therefore, presumably, at least somewhat less likely to expand its international (and domestic) spending.

Would these contrary influences be fully offsetting? No. Forty-six years ago Hicks [1935, p. 10] pointed out: "borrowing is nearly always a voluntary act on the part of the borrower, which would not be undertaken unless he was willing to become less liquid than before. . . .On the other hand, if the borrower's credit is good, the liquidity of the lender will not be very greatly impaired by his making the loan. . . Thus the net effect of the loan is likely to be 'inflationary,' in the sense that the purchase of capital goods or securities by the borrower is likely to be a more important affair than any sale of capital goods or securities by the lender, made necessary in order for the lender to restore his liquidity position."

Hence case (b) would be *somewhat* more expansionary than (a). Note also that the expansionary influence operates without any increase in the world's money stock, and indeed with some overall reduction in world liquidity relative to the higher level of world expenditure and receipts.

Nevertheless, it is most unlikely that such credit could contribute greatly to world inflation. The deficit country suffers the illiquidity of increased debt. The surplus country holds a financial asset, but it is a very "illiquid" one. The lending which financed the expansion is likely fairly soon to bring about increased "saving," either by Country A as it services or pays down its debt, or by Country B. For B's expanded activity would require additional credit, which would be difficult to

secure, except from "saving," in the absence of financial intermediation. And if the source of funds for lending becomes true saving, it is no longer expansionary. There are also special risks, to one party or another, if the debts are denominated in the other's currency. The reduced monetary liquidity may even play a role.

In sum, where as here the world's credit system is so primitive, and so rigidly tied to an inelastic "monetary base," the constraints of increased debt and debt service on the borrowers, combined with the growing illiquidity and increased credit risk of the lenders, would before long bring the credit expansion that fueled the expanded spending to an end. There might be short-term "credit cycles," but not persistent inflation. In such a world, money matters very much, and the monetarists would have the weight of sound economics on their side.

III. Simple (Monetary) Financial Intermediation

The model used in Table 1 is, of course, more a caricature than a facsimile of the world's financial system even before the recent developments that I wish to stress. Way back before World War I most developed nations had financial intermediaries, and many of them had central banks. If only because of domestic banking systems, the linkage between credit and international money (e.g., gold) had taken on accordian properties.

The way in which this simple intermediation increases expansionary potentialities is examined in Table 2. Each of our Countries A and B now have banking systems, which finance their lending by borrowing solely in demand deposit form. It is assumed for simplicity that there is no international lending, between either banks or non-financial units, and that foreign claims on each banking system are settled by direct foreign exchange transactions. We assume initially that the two countries' banking systems "keep in step" in expanding imports financed by increased bank credit.

The upshot is shown on line (Σ) of the table[1]. In effect the exporters of each country finance their importers by holding demand deposit liabilities of banks. Note again that the importers lose no cash liquidity. The exporters, on their side, again enjoy increased export earnings but now they hold increased financial assets in a presumptively safe and

[1] It is assumed in the table that each banking system directly credits demand deposit accounts in making its loans (line (1)). But much the same results would occur if "acceptance credit" were used. In that case the demand deposits would emerge as the exporters sold their importers' drafts to their banks in exchange for local currency demand deposits. The imports and exports and the exchange of checks or drafts are shown on line (2). The foreign exchange transactions are shown on line (3). Line (Σ) sums the transactions on lines (1) to (3), eliminating offsetting transactions.

Table 2 — *Simple (Monetary) Financial Intermediation*
(the balanced case)

	Country A				Country B				World	
	non-financial sectors		banking system		non-financial sectors		banking system		non-financial economy	
	uses	sources	uses	sources	uses	sources	uses	sources	uses	sources
(1)	+ DD_A	+ loanL	+ loans	+ DDL_A	+ DD_B	+ loanL	+ loans	+ DDL_B	+ DD	+ loanL
(2)	+ imports (E)	− DD_A			+ DD_A	+ exports (R)				
	+ DD_B	+ exports (R)	− DDL_A	+ DDL_A	+ imports (E)	− DDL_B	− DDL_B	+ DDL_B	+ imports (E)	+ exports (R)
(3)	+ DD_A	− DD_B	− DDL_A	+ DDL_A	+ DD_B	DD_A	− DDL_B	+ DDL_B		
(Σ)	+ DD_A + imports (E)	+ loanL + exports (R)	+ loans	+ DDL_A	+ DD_B + imports (E)	+ loanL + exports (R)	+ loans	+ DDL_B	+ DD + imports (E)	+ loanL + exports (R)
(x)	− E_{Dom}	− R_{Dom}	—	—	− E_{Dom}	− R_{Dom}			− E_{Dom}	− R_{Dom}

Legend: L = liability; DD = demand deposits; E and R = expenditures and receipts on goods and services; Dom = domestic. The subscripts A and B designate each country's unit of account.

highly liquid form. The exporters can now finance increased spending without either saving (i.e., spending less than their receipts) or securing credit. In all likelihood it will require still further increased spending, domestic and/or international, before the surpluses which must match the deficit spending in order to bring about expenditure equilibrium, develop[1].

Potential expansion is clearly much enhanced by this simple intermediation. Opinions may well differ whether credit or increased liquidity is more responsible for this enhancement. The initial deficit spending of the importers must clearly be attributed to credit, since credit finances their spending and they end up with no more money and somewhat decreased overall liquidity. But whether the induced increased spending of the exporters (and other domestic non-financial units) is more influenced by their increased sales and incomes or their enhanced liquidity will be judged differently by Monetarists and by others. But the crucial matter, in my view, is the much greater willingness to *lend* because of the greater safety and liquidity of the banks' obligations than those of the non-

[1] There is of course a *possibility* that the increased foreign spending of the importers of A and B would be directly offset by reduced domestic spending in the two countries, i.e., by concurrent saving. This possibility is indicated on line (x) of the table by the reduced E_{Dom} and R_{Dom}. In this case total world spending would not rise. This, however, is extremely unlikely where, as here, the financing of the deficits is via the increased demand deposits of banks. The exporters are far more likely to increase their spending than to decrease it. In fact the "expenditure multiplier" is likely to be quite substantial.

financial borrowers whose deficit spending is being financed. This leads towards further expansion of lending to finance deficits. Note also that the lending can be largely independent of *concurrent* saving.

There are nevertheless several important financial constraints upon inflationary expansion in this model. One, likely very minor in practice, is that non-financial units may wish to hold increased amounts of "hard money" or legal tender as their spending rises. More potent are three other constraints. There is first a good possibility that the banking systems do not "keep in step," so that the one that "takes the lead" will lose international reserves and be forced to curtail credit. The second is that because of legal or conventional reserve requirements, the banking systems may have to cease expanding. A third constraint may be imposed by national central banks. By restrictive open market policy they may be able to reduce the reserves of their banking systems. It is the weakening of each of these three constraints that, in my judgment, mainly explains the world's credit explosion and the consequent worldwide inflation of recent years.

IV. The Dollar Currency System

Chronologically the first important development leading towards inflationary international finance in the post-World War II years was the growth of the so-called Dollar Currency System under the Bretton Woods agreements. Under this system the dollar liabilities of US banks became an almost universally used Vehicle and Reserve currency.

It should first be noted that the international use of a common currency is not, in itself, especially expansionary. While the system undoubtedly facilitates international trading by providing a more efficient unit of account and means of payment, using dollar accounts for these purposes will have only moderate influence on world spending unless the volume of dollar lending and liabilities is progressively increased. If international transactions among dollar users are balanced, the debits and credits to every nation's accounts with US banks are equal and offsetting. If one country runs a deficit, this simply shifts dollar funds from that nation to others. In the absence of credit expansion, in neither the balanced nor the unbalanced cases is world spending substantially expanded, since there is neither credit-induced spending nor increases in worldwide liquidity.

Where the system is combined with progressive credit expansion, however, it is strongly expansionary. This is analyzed in Table 3. It is assumed at this initial stage - - what actually occurred during some parts of the worldwide inflation - that the United States runs an inter-

Table 3 — *US Banks Finance a US Import Deficit under the Dollar Currency System*

	United States (US)				Foreign countries (F)					
	non-financial sectors (NFs)		commercial banks		non-financial sectors (NFs)		private banks		central banks (Cbs)	
	uses	sources	uses	sources	uses	sources	uses	sources	uses	sources
(1)	+ $DD	+ $loanL	+ $loans	+ $DDL	--					
(2)	+ imports (E)	— $DD	— $DDL to US NFs	+ $DDL to F NFs	+ $DD	+ exports (R)				
(3)	—		— $DDL to F NFs	+ $DDL to F banks	+ FDep	— $DD	+ $DD	+ FDepL	-	
(4)	---		— $DDL to F banks	+ $DDL to F CBs	--		+ FRes	— $DD	+ $DD	+ FResL
(Σ)	+ imports (E)	+ $loanL	+ $loans	+ $DDL to F banks or CBs	+ FDep	+ exports (R)	+ $DD or + FRes	+ FDepL	+ $DD	+ FResL

Legend: $DD -- US dollar demand deposits; F = Foreign or foreign currency; Dep -- demand or other types of deposits with foreign banks; Res = bank reserves; NF = non-financial units or sectors; L - liabilities; E - expenditures; R = receipts.

national trade deficit, financed by demand deposit loans to US borrowers by US banks.

On line (1) US banks lend to US non-financial sectors. This leads to increased US imports on line (2). On the foreign side, two alternative uses of the added dollar funds received from exports are shown. On line (3) the exporters sell their increased dollars to private banks for local currencies (in effect *lending* the funds to their banks in deposit form)[1]. The foreign banks may use these dollars to do dollar lending, but alternatively they may sell the dollars to their central banks in exchange for foreign currency bank reserves, as on line (4). (In effect, they are lending the funds to the central banks.) In that case, it is assumed (provisionally) that the central banks hold (lend) the dollar funds as demand deposits with US commercial banks.

The bottom line (Σ) summarizes the net effects of these transactions on each set of parties involved. The first important thing to note is that the increased dollar lending and spending lead to no loss of US bank reserves. On each of lines (2) to (4) US banks simply debit some dollar accounts and credit others. The US and its banking system need no longer "keep in step" with others. This in itself is expansionary in tendency.

[1] Note that *demand* deposits are not specified. The funds may be placed in other deposit forms.

The other important effects occur in the foreign countries. Foreign non-financial sectors have increased export earnings and also increased local currency deposits. Both income and liquidity effects are positive. Moreover, and this important point is too often ignored, *foreign banking systems have increased dollar or foreign currency reserves to lend.*

How much of the expanded spending explicit and implicit in Table 3 should be ascribed to credit expansion and to increased liquidity, respectively? Certainly the direct expansion of US imports and exports must be primarily ascribed to the former. Both the US non-financial sectors and banks become somewhat less rather than more liquid. The increased money stock of dollars is held by foreign private or central banks.

On the foreign side, however, there are expansionary income, liquidity, and credit effects. The non-financial sectors secure expanded income and also increased liquidity. The foreign private banks secure increased liquid reserves, which encourage them to expand their lending. The increased spending of their new borrowers must again be ascribed to credit rather than liquidity effects, but it is the increased liquidity of the private banks that induces this lending. We see here a second example of the important principle: *the increased liquidity that leads to expanded spending operates primarily through the liquidity of lenders, and especially of financial intermediaries, rather than through the liquidity of those who spend directly in the goods and services markets.* The increased liquidity of the foreign non-financial sectors may do something to induce further spending, but the strategic increased liquidity is that of the foreign banks (or other financial institutions) which secure loan funds that encourage them to do further lending[1].

The stimulus that US bank lending imparts to foreign credit expansion and world spending under the Dollar Currency System does not depend, as is often thought, on a US trade deficit. This is seen when US banks lend to foreign borrowers who spend the dollar proceeds on imports from other foreign countries. Here the deficits which lead to increased spending occur in foreign countries. The US banking system acts essentially as an international financial intermediary, as Despres *et al.* [1966] have pointed out[2].

[1] It will be noted that the US money supply, as it is now defined by the Federal Reserve System, is not increased by the above transactions. The increased demand deposits are held by foreign financial institutions. The most significant financial effect, so far as the US is concerned, is the decreased liquidity of the US banking system as its loans and demand deposit liabilities rise. How this is combatted is dealt with in the next section.

[2] Indeed it is quite possible that US bank lending to foreign borrowers may be conducive to a US trade surplus if the dollar proceeds are spent on US exports.

It is of course conceivable that the direct and indirect stimuli to world spending shown in Table 3 would be negated by concurrent saving. This, however, is extremely unlikely. There is relatively little except increased money income that would lead to increased saving in the processes examined. The lending transactions have little if any of their sources in saving. They are mainly transfers of available funds to more remunerative uses than holding excess bank reserves or idle money balances. A large "expenditure multiplier" would therefore be expected.

V. The Growth of Liability Management

The dollar loan expansion upon which much of the expansionary influence of the Dollar Currency System depended led, as seen, to a weakening of the reserve position of the US banking system. This particular linkage between "money" (bank reserves) and world credit extension, while weakened, still potentially existed.

An important development of the 1960s and 1970s much further weakened this linkage, increasing both US and world credit availability and extension. This was the growth of "Liability Management." Aggressive Liability Management came particularly to characterize the behavior of the large US commercial banks heavily engaged in international finance. The Eurocurrency banks, dealt with in the next section, were also built on Liability Management[1].

US bank Liability Management involved notably the sale of negotiable certificates of deposit (CDs), and operations in the Federal Funds market. The essential way in which it furthered US bank credit expansion can be conveniently illustrated by analyzing the sale of CDs by US banks to

[1] Begun largely as a method of adjusting temporary reserve positions by borrowing or lending short term funds rather than selling or purchasing short term assets, Liability Management gradually developed into the large-scale and aggressive borrowing of market sensitive funds for permanent bank financing and bank growth. Its use for merely adjusting temporary reserve positions has relatively negligible implications, but, as fully developed, Liability Management represents a fundamental change in the character of commercial banking. It makes obsolete the traditional models of a banking system reacting mechanically to changes in bank reserves or a "monetary base" via money or credit multipliers. For fuller treatment of bank Liability Management see Earley and Thompson [1977].

The growth of Liability Management has not been confined to commercial banks. It characterizes the present operations of important non-bank financial intermediaries in the United States, including the large Savings and Loan Associations. It has also helped large non-financial businesses operate with much lower ratios of cash and liquid assets to outlays and liabilities, making them more dependent on credit lines and loan commitments. Here again the links between money and spending (and saving) were greatly weakened. The working balances of non-bank financial intermediaries and large non-financial firms have been a major source of loan funds supporting bank Liability Management, by purchase of certificates of deposit and the sale of Federal Funds.

Table 4 — *Liability Management via Bank CD Sales Relieves US Bank Reserve Pressure, Permits Further US Bank Lending, and Encourages Foreign Bank Lending and World Spending*

	US banking system		CD purchasers		World NF sectors		Foreign banks			
							private		central	
	uses	sources	uses	sources	uses	sources	uses	sources	uses	sources
(1)	− $DDL	+ $CDL	+ $CD	$DD						
(2)	+ $loans	+ $DDL			+ $DD	+ $loanL				
(3)	− $DDL	+ $DDL			− imports (E)	+ exports (R)				
(4)	− $DDL	+ $DDL			+ FDep	− $DD	+ $DD	+ FDepL		
(5)	— $DDL	+ $DDL					+ FRes	— $DD	+ $DD	+ FResL
(Σ)	+ $loans	+ $CDL	+ $CD	$DD	+ FDep + imports (E)	+ $loanL + exports (R)	+ $DD or + FRes	+ FDepL	+ $DD	+ FResL

Legend: See Table 3. CD = Negotiable certificates of deposit.

either domestic or foreign holders of US demand deposits. This is done in Table 4.[1]

On line (1) the CD purchasers draw down their demand deposit balances to purchase CDs from banks desiring to expand their lending. For the US banking system as a whole the net result is simply the conversion of demand deposit liabilities into CD liabilities. Since reserve requirements on CDs have been very substantially lower than on demand deposits, this conversion creates excess reserves for the US banking system.

The excess reserves acquired by the CD selling banks enable them to expand their loans to foreign or domestic borrowers (line (2)). In either case, under the Dollar Currency System, the US banking system loses no reserves, although the original level of demand deposit liabilities is (substantially) restored. The upshot, shown on line (Σ), is US bank loan expansion, increased imports and exports, increased foreign bank deposits, and excess reserves for the foreign banking systems[2].

It will be noted that these operations do not increase the US money supply, as conventionally defined. In fact there is an initial fall in the US M1s and in M2, which are restored only as the banks expand their loans. But the financial deficits that accompany the expanded lending and spending necessarily bring financial surpluses to other parties,

[1] For further treatment see Earley and Thompson [1977].

[2] On line (4) it is assumed, as in Table 3, that exporters convert their dollar proceeds into local currency deposits and on line (5) that the foreign private banks sell their dollars to their central banks for additional reserve funds. Provisionally, it is assumed that the foreign central banks hold the US bank demand deposits.

somewhere in the world. And if, as is highly likely, these surpluses lead to flows of funds to foreign financial intermediaries, there is the same impetus to still further credit expansion and spending that we observed in Table 3.

Whether the CD funds the banks borrow are secured from US or foreign non-financial sectors, from foreign private banks or central banks, or from US or foreign non-bank financial institutions, world credit is almost certainly expanded[1]. Stated generally, if the CD purchasers do not have to reduce fully commensurately either their spending or their lending to other parties, there will be net world credit expansion and higher world expenditures. This is almost certainly the case. For the freeing of US bank reserves brought about by the conversion of demand deposit liabilities to CDs does not significantly decrease the "liquid reserves" of either spenders or non-bank lenders. It simply converts them into highly liquid earning assets rather than demand deposits. And the resulting bank loan expansion in fact restores dollar demand deposits, so that the world money stock as usually defined is not depleted. In the event, it simply grew much more slowly than world credit[2].

VI. The Eurodollar Markets

I move now to the expansionary influence of the Eurodollar markets, which developed, quite naturally, as an adjunct to the Dollar Currency System[3]. The many conditions that contributed to the phenomenal growth of Eurodollar banking have been well covered in the voluminous literature. For present purposes it is enough to note only a few. There is the obvious advantage of a substantial interest yield in holding dollar funds with Eurodollar banks rather than as demand deposits with US banks. To the Eurodollar banks, important advantages include an absence of reserve requirements and freedom from interest rate regulation. To the

[1] Only if, directly or indirectly, funds are withdrawn from the Eurodollar markets to purchase the CDs would world credit conditions tighten significantly; this will be examined later.

[2] Liability Management is not of course solely connected with the international side of US bank operations. The buyers of CDs may be domestic or overseas holders of US bank deposits, and the borrowers who secure loans as a result of the freeing of reserves may be foreign or US borrowers, whose loan-expenditure may be in US, foreign country, or international markets.

[3] In dealing with the Eurodollar markets it becomes advisable to shift attention from national deficits and surpluses, and adopt instead the alternative focus which Salant, in Krause and Salant [1977], has recommended for analyzing worldwide inflation, namely that of the worldwide financial and trading system.

borrowers from Eurodollar banks advantages may include lower interest costs, greater "availability," and preferred maturities or other terms.

Eurodollars arise from the transfer of demand deposit or other liability accounts of US resident banks to liability accounts of banks or banking offices located in other countries. The latter then hold the US bank funds, pending their being lent to other banks or non-bank borrowers. Under the Dollar Currency System or its successor (e.g., floating dollar exchange rates) the resident US bank dollars "never leave home," being simply transferred from one party to another. But the volume of Eurodollar accounts can grow progressively with transfers from resident banks to Eurodollar banks. Eurodollar accounts cannot be drawn upon by check or draft, but the bulk of them are quite short in maturity, varying from "sight" to several months. Substantial amounts are held as negotiable certificates of deposit. Most of the liabilities of Eurodollar banks are correctly perceived as highly liquid[1].

If the source of the funds transferred into Eurodollars is simply US resident bank demand deposit liabilities, the expansionary influences of the Eurodollar markets are quite straightforward, as in Table 5. Essentially, the Eurodollar banks secure dollar funds to lend without any decrease in the reserves, nor increase in the reserve requirements, of the US banking system, and hence no need for it to contract credit. As a result, world bank credit is eased, world "credit availability" increases. Given a strong demand for credit, world credit expands, and world spending rises[2].

[1] The concept of the Eurodollar has been unusually elusive and variegated, ranging between the bizarre extremes of "greenbacks circulating abroad" (newspapers) and overseas dollar bank accounts created by "a stroke of the bookkeeper's pen" [Friedman, 1969]. As is usual, confusion in concept has led to confusion and error in conclusions. The concept used here stresses the essential borrowing and lending nature of Eurodollar transactions, and the close relation between Eurodollar transactions and the Dollar Currency System. The most complete and satisfactory treatment of the nature of Eurodollars and the Eurodollar markets is probably that by Einzig [1964; *idem* and Quinn, 1977], whose six editions spanned the period from 1964 to posthumous 1977.

[2] The numerous views disputing the conclusions expressed in this and the following section can usefully (although not sharply) be divided into those that would dispute the above-stated results of simple demand deposit transfers and those which dispute the more complicated conclusions of Section VII.

The contentions that mere transfers of demand deposits are not significantly expansionary usually rest on one or more of three grounds. The first is that Eurodollars are not a means of payment and their increase does not, therefore, increase the world "money supply" [Mayer, 1971; Throop, 1979]. The second is that such transfers correspondingly reduce the credit-extending capacity of the US banking system [Mayer, 1971; Wood and Mudd, 1977]. The most common source of disagreement is the conclusion, based in a number of cases on painstaking empirical research, that unlike that of a closed banking system, the money and/or credit "multiplier" for Eurodollar banking is very small, perhaps slightly above one, but more likely considerably less. From the (correct) position that an injection of funds into

Table 5 · *The Influence of Transfers of Demand Deposit Balances at US Banks to Eurodollar Banks*

		World non-financial sectors		US banks		Eurodollar banks		US and Eurodollar banks combined	
		uses	sources	uses	sources	uses	sources	uses	sources
1	(a)	+ E$D	$DD	− $DDL	+ $DDL	+ $DD	+ E$DL	+ $DD	+ E$DL
	(b)	+ $DD	+ E$loanL	− $DDL	+ $DDL	+ E$loan	$DD	+ E$loan	− $DD
	(Σ)	+ E$D	+ E$loanL	0		+ E$loan	+ E$DL	+ E$loan	+ E$DL
2	(a)	+ E	+ R	− $DDL	+ $DDL	0		0	
	(b)	+ Ė	+ Ṙ	$DDL	+ $DDL				
3	(a)	0		0		0		0	
	(b)	+ $CD	$DD	$DDL	+ $CDL			$DDL	+ $CDL
	(c)	+ E$D	− $DD	$DDL	+ $DDL	+ $DD	+ E$DL	+ $DD	+ E$DL

Legend: $ = dollar funds with US banks; E$ = dollar funds with Eurodollar banks; DD = demand deposits; D = non-demand deposit funds with Eurodollar banks; L = liabilities; E = expenditures in the goods and services markets; R = receipts from these expenditures; Ė and Ṙ = induced E and R via multipliers and accelerators.

This case is analyzed in Table 5. In part 1 demand deposit funds held by non-financial units are lent to Eurodollar banks. This may well shift reserves among US banks, but the US banking system as a whole loses neither reserves nor demand deposits, whereas the Eurobanks gain loanable funds. On line 1 (b) the Eurodollar banks lend these funds to non-financial borrowers[1]. The net effects to this point are set forth on line 1 (Σ).

Part 2 of the table illustrates the effects of these transactions on world spending on goods and services. Probable added spending and receipts via expenditure "multiplier" and "accelerator" effects are indicated by Ė and Ṙ on line 2 (b).

Let us also examine some of the *financial* repercussions of this added spending. Under normal conditions, portions of the resultant added receipts will be "saved," constituting increased "financial surpluses" in our terminology. This saving will, of course, tend to limit world income expansion. If we were strictly "Keynesian" we would go no further. But as Keynes himself reminded the economics profession, a micro-unit has also to make choices as to how to *use* the funds it chooses, even temporarily,

the Euromarkets does not lead to "automatic" multiple expansion, the (incorrect) inference is drawn that the Euromarkets cannot therefore be significantly expansionary. For fuller treatment, see Section VII below.

[1] Much of the lending of Eurodollar banks is to other banks; the clearly expansionary lending is that to borrowers who spend the funds in the goods and services markets. Interbank lending is important mainly because it provides the banks with great liquidity, via this form of Liability Management. It thus encourages expansionary lending. Interbank lending is also heavily involved in Eurobank foreign exchange dealings.

not to spend. The repercussions on credit conditions, lending and spending depend much on what is chosen.

In section 3 of Table 5 we analyze those three among the various choices which are most relevant to the influence of the combined Dollar Currency System and Eurodollar markets on worldwide spending and possible inflation. These are (a) to hold the dollar receipts as demand deposits with US banks, (b) to make time-dated loans to them (e.g., to buy newly-issued US bank CDs), and (c) to invest the funds with Eurodollar banks.

Line 3 (a) depicts the first of these choices. There is no further activity involving US banks and hence no change in the lending power of the US banking system. In a now almost forgotten phrase of Keynes, these "savings" will have "run to waste" — which would be all to the good in an inflationary milieu.

If instead, as on line 3 (b), the non-financial recipients use their demand deposits to purchase newly-issued US bank CDs, this frees US bank reserves and gives US banks added funds to lend. The expansionary repercussions of such transfers have already been traced in Table 4.

On line 3 (c) it is assumed that at least part of the dollar demand deposit funds received by non-financial recipients of the spending on line 2 is transferred to Eurodollar banks. In this case the Eurodollar banks secure added loan funds, again without depriving US banks of reserves, and world credit will tend to expand further.

It must be emphasized that no automatic or mechanical "Eurodollar multiplier" is postulated. The proportion of the total newly created loan funds that arise out of added world spending cannot be mechanically specified, and the fraction of those which will flow to Eurodollar markets depend mainly upon competitive factors in the world credit markets. It is simply assumed that larger world money receipts are likely to lead to some further lending through this relatively attractive and aggressively operated segment of the world's credit system[1]. We will deal in Section VII with the competitive influences of the choice between lending in US and Eurodollar markets.

[1] The analysis assumes that in all three cases in section 3 of Table 5 the funds are "saved" in the basic sense they do not go back into goods and services markets. It might appear therefore that the processes depicted in 3(b) and 3(c) are not expansionary, since if increased credit and spending only offset concurrent saving, aggregate spending will be unaffected. This is incorrect. If such further lending puts otherwise "idle funds" into goods and services markets the constraint on aggregate spending brought about by saving will be to that extent eased. It is the general fact that funds for lending so often do *not* have their source in saving that makes financial intermediation so expansionary. The immense volume of liquid funds, plus the aggressive efforts of intermediaries to borrow them, is what gives a strong inflationary bias to the contemporary financial system.

Let us summarize to this point the influences of the Eurodollar markets on world liquidity, credit extension, and spending. The initial transfer of demand deposit funds to Eurodollar banks slightly reduces the liquidity of the world non-financial sectors. The subsequent lending by the Eurodollar banks restores world non-financial sector money holdings, but only (roughly) to their original level. Meanwhile the expanded level of world money expenditure further reduces world monetary liquidity relative to world spending. If a portion of the expanded receipts are transferred to US bank CDs, there is some further slight reduction in the liquidity of the non-financial sectors, and broadly similar liquidity effects flow from any lending of demand deposit funds in the Eurodollar markets.

The only clear increases in liquidity in the entire set of transactions occur: (1) when the Eurodollar banks secure added loan funds and (2) when the US banks enjoy an easier reserve position because of a reduction in their required reserves. These "transient" improvements in the liquidity of these institutions, however, encourage them to increase their lending, and this raises world expenditure. It is clear that the prime financial cause of the rising world spending is the extension of credit via financial intermediation rather than any increased liquidity of the non-financial sectors[1].

It is important to note, however, that — except for non-financial borrowers and the US and Eurodollar banks — the decline in world liquidity is rather slight. Large parts of the liabilities of Eurodollar banks have short maturities, and the credit standing of most Eurodollar banks is very strong. The negotiable certificates of deposit of US banks are also highly liquid and presumptively very safe. Hence the impetus to world spending provided by the transfer of dollar demand deposits to Eurodollar market lending is only very mildly constrained by growing world illiquidity. In any case the linkage between world money supplies on the one side and credit and world spending on the other is further weakened by these operations in the Eurodollar markets.

VII. Eurodollar Funds from Non-Demand Deposit Sources

Once US bank financing had become heavily dependent on non-demand deposit sources, Table 5 ceased to be the typical case. Although

[1] It is true, of course, that the world non-financial sectors have increased liquid assets in the form of Eurodollar deposits and/or negotiable dollar CDs. To some extent, perhaps, the possession of these "close money substitutes" may lead to higher propensities to spend rather than run financial surpluses. But this is at most a secondary effect. Much more likely is it that possession of these liquid assets will strengthen the disposition of non-financial sectors to lend more demand deposit funds in non-money forms to banks and Eurobanks. The major expansionary impulses via liquidity, in short, arise in the financial rather than the goods and services markets.

the accounting transfer to the Eurodollar banks would (normally) be from demand deposit accounts, the indirect sources of these funds would very often be the sale or liquidation of either bank or non-bank short-term time debts denominated in dollars.

Where the source is the sale of US open market securities (e.g., Treasury bills or commercial paper) the consequences would be only slightly different from those already analyzed. Notably, the reserve position of the US banking system would not be weakened. The buyers of the securities would transfer existing demand deposit balances to the sellers, who would have them transferred to Eurodollar banks. Again the latter would secure loan funds without any loss of loan funds available to the US banks[1].

The situation is much more intricate where US banks and the Eurobanks are directly competing for interest-sensitive funds — as, for example, when the US banks are financing much of their loan operations by issuing negotiable CDs. Table 6 analyzes this case.

Panel I of the table shows the transfer of CD funds to Eurobanks. The US banking system experiences a tightened reserve position because of the conversion of CD to demand deposit liabilities, with their distinctly higher reserve requirements. The extent, if any, to which the US banks must contract credit, and the overall effects upon world credit conditions, depend partly on "mechanical" differences in reserve practices, but much more upon the manner in which the US banks react to their reserve deficiency.

Even if the US banks reacted passively by reducing their loans, as in panel II.A, the amount by which their loans would have to be reduced would be considerably less than the amount by which the Eurodollar banks could expand theirs, owing to the difference between US reserve requirements on demand deposit liabilities and the very small dollar reserves actually held by Eurodollar banks[2]. Nevertheless, a substantial shift of

[1] The main effect, so far as the US side is concerned, would be to tighten short-term open market credit conditions somewhat. New borrowers in these markets would have to pay somewhat higher interest costs. Since the borrowers that typically finance their operations by US open market paper — i.e., the US Treasury, large non-financial corporations and large finance companies — would be quite unlikely to reduce their spending (or lending) significantly because of slightly higher short-term borrowing costs, the net effect, when the Eurobanks had lent the transferred funds, would be that worldwide credit, and spending, would rise.

[2] If, for example, US reserve requirements on demand deposits were 15%, and on CDs 5%, and \$ 1 billion of CD funds were run off and transferred to Eurobanks, the US banking system would initially have a reserve "deficiency" of \$ 100 million. US banks would have to reduce their loans enough to reduce their demand deposit liabilities by \$ 667 million (\$ 100 million ÷ .15) to restore their former reserve position. This reduction can be compared with the almost \$ 1 billion of loans which the Eurobanks could add to their portfolios, given their practice of keeping extremely small balances with US banks.

Table 6 – – *US Banks and Eurodollar Banks Compete for Interest-Sensitive Funds: US Bank CD Funds Are Transferred to Eurodollar Banks and US Banks Make Alternative Adjustments*

		Other world sectors uses	Other world sectors sources	US banking system uses	US banking system sources	Eurodollar banks uses	Eurodollar banks sources
I. CD funds are transferred to Euro-$ banks	(1)	+ E$D	– – $CD	– – $CDL	+ $DDL	+ $DD	+ E$DL
	(2)	+ $DD	+ E$loanL	– – $DDL	+ $DDL	+ E$loan	– $DD
	(Σ)	as above		– $CDL	+ $DDL	+ E$loan	+ E$DL
II. Alternative adjustments: A. US banks reduce loans		– – $loanL	– – $DD	– $DDL	– $loan		
	(ΣΣ)	+ E$D $loanL	+ E$loanL	– $CDL	$loan	+ E$loan	+ E$DL
B. US banks sell securities		+ $Sec	$DD	– $DDL	$Sec		
	(ΣΣ)	+ E$D + $Sec	$CD $DD	– $CDL	$Sec	+ E$loan	E$DL
C. US banks replace CDs		+ $CD	– – $DD	– $DDL	+ $CDL		
	(ΣΣ)	+ E$D	+ E$loanL	0		+ E$loan	+ E$DL
D. US banks buy Fed Funds		+ FedF	$DD	– $DDL	+ FedFL		
	(ΣΣ)	+ E$D + FedF	+ E$loanL – $CD	– – $CDL	+ FedFL	+ E$loan	+ E$DL

Legend: See earlier tables. Sec – short-term open market securities; FedF – Federal Funds sold; FedFL = Federal Funds liability (Fed Funds bought); the lines (ΣΣ) show the combined effects of section I and the adjustments made in section II.

funds to Eurodollar banks would cause some disruption in US bank loan markets. The banks would have solved their reserve problem, but neither they nor their loan customers would like it. It is therefore most unlikely that the banks would choose this solution.

Section II. B shows the sources and uses of funds if the banks instead reacted in their once traditional manner, that is by selling open market securities from their portfolios to non-bank units holding demand deposits. Sales of such securities equal in amount to their loss of CDs would restore the banks' reserve position.

This solution would be less disturbing to US credit markets than the calling of bank loans. Nevertheless the non-bank sectors would be made appreciably less liquid by this substitution of a combination of Eurodollar deposits and US open market securities for their former holdings of demand deposits and US bank CDs. In order to induce this substitution on any large scale, US short-term open market interest rates might have to rise fairly substantially[1].

[1] Three facts, however, make it virtually certain that the credit expansion in the Eurodollar markets will much exceed any contraction of credit that may occur in US markets. The first is that under prevailing reserve practices the amount of securities that would have to be sold by US banks would be considerably less than the expansion of Eurodollar loans. The second is that the higher short-term interest rates that would normally accompany the transfer of funds from US time obligations to Eurobanks would strengthen the inducement to shift funds from demand deposits to bank time deposits and CDs, thus easing the banks' reserve position and reducing the amounts of securities they would have to sell. The third fact is that those who use US short-term open market paper for their financing are unquestionably much less sensitive to short-term credit costs than non-financial borrowers from Eurobanks are to Eurobank "credit availability."

In any case, this traditional manner of solving bank reserve deficiencies would be an unlikely reaction in the present era of Liability Management. Aggressive banks would be quite unhappy about the resulting decline in their gross size, earning assets, and (probably) net earnings. Hence they would undoubtedly instead seek to replace their reserve funds by new borrowing in the CD or Federal Funds markets.

Section C shows the results if the banks simply replace their maturing CDs with new ones. The increased demand deposits shown in non-bank sector hands in panel I now disappear, being exchanged for new CDs. The amount and composition of US banking system assets and liabilities would be unchanged. The US banks would not have to contract their loans, but the Eurodollar banks will have expanded theirs. Thus worldwide credit would have expanded[1].

Another feasible, and indeed very likely, solution to the reserve problem of the US banks is shown in panel D. Here the US banks protect their reserve positions by net purchases of Federal Funds. They buy these Fed Funds not from other Member banks, of course, but from those parties which in fact sell the bulk of net Fed Funds sold, namely Non-Member banks, non-bank financial institutions, and the banks' large corporate customers. As Earley and Thompson [1977] have shown, the net effect of such transactions is for the sellers of the Fed Funds to exchange their demand deposit balances for the highly liquid earning asset Fed Funds sold. The US member banking system as a whole shows increased Fed Funds purchased in place of demand deposit liabilities. As shown on the bottom line of the table, the banks will in effect have substituted Fed Funds liabilities for their extinguished CD liabilities.

[1] There will be some upward pressure on interest rates, it is true, but it is quite unlikely that the higher rates will seriously disrupt credit markets. The funds that are made available to credit markets come from several sources. There is first the conversion into the time obligations of banks and other financial intermediaries of bank demand deposit liabilities. With interest rate ceilings not applying to short-term CD borrowing by the banks or to borrowing by the Eurobanks, but applying to US demand deposits, only moderate rises in open market rates will induce much transference of demand deposit balances to CDs or Eurobank deposits. In addition, as explained earlier, the expanded receipts and incomes generated by the US bank and Eurobank lending will themselves in due course tend strongly to create more loan funds available to provide the bank CD replacement.

Within limits, this is the easiest, most flexible, cheapest and preferred solution of the US banks' reserve problem as they face the competition for time funds from the Eurodollar markets. Although the probable resulting rise in the Fed Funds rate will tend to raise other short-term interest rates, the same facts noted above assure that world credit will have expanded by the transfer of bank time funds to the Eurodollar markets[1].

Eurobanks practice Liability Management even more intensely than resident US banks. They normally keep only negligible demand deposits with US or other banks. They solve their liquidity problem in part by some matching of the maturities of their obligations and claims, but mainly by borrowing needed replacement funds, either in the Eurodollar markets or from US banks. The fact that many Eurodollar banks are branches or offices of US banks assists them to secure the requisite funds. US bank lending through their branches to the Eurodollar markets is a common practice.

Hence the US banking system and the Eurodollar markets represent a fairly closely integrated system of credit markets. Eurobank operations are very sensitive to US credit conditions, and to a lesser degree US credit conditions are sensitive to conditions in the Eurodollar markets[2]. But

[1] It is of interest to trace the effects of the transactions in Table 6 on the world's narrowly defined money supply. An increase does temporarily occur in panel I of Table 6, but each of the US bank adjustments erases this increase. Since no increase in the world "money stock" actually occurs, the increased spending resulting from the Eurodollar lending cannot be attributed to this particular "liquidity effect." It is rather the increased lending capacity of the Eurobanks, combined with the US bank adjustments that permit them to maintain their own lending, that accounts for the credit expansion and the growth of world spending.

[2] It is well to point out a basic if only partial asymmetry between the position of the US banking system and this important "satellite" of the Dollar Currency System. In the absence of gold convertibility or conversion, if the Eurobanks manage to attract funds from US credit markets, the US banking system loses no reserves. Even if a transfer of bank time deposits or CDs tightens the reserve position of the US banks by changing the composition of their liabilities, a relatively simple solution lies at hand, namely to borrow enough new funds in the CD or Fed Funds markets, or directly from the Fed, to restore the previous liability composition. Bank interest costs may rise, but there need be no "disintermediation," and hence no crippling effect on bank "credit availability" in the United States.

The situation is different when conditions in the US credit markets threaten withdrawal of funds from the Eurodollar markets. In effect, the Eurobanks are threatened with the loss of their "reserves," of which they have almost none. They have no direct access to Central Bank credit. They have either to liquidate loans to the full amount of the funds withdrawn or borrow that full amount from other sources. Since these sources are largely the same ones wishing to shift funds to US banks and US markets, the Eurodollar banks are faced with a much more formidable liquidity problem. Interest rates in Eurodollar markets may have to rise very high, and Eurodollar "credit availability" may shrink seriously, before this liquidity problem is solved. One solution, of course, is to borrow, directly or indirectly, from US banks, which in turn may borrow from CD or Fed Funds markets or directly from Federal Reserve Banks. This is commonly done.

for reasons already explained, it is incorrect to see them as a fully integrated market merely allocating a given volume of funds, as do some authorities. Essentially, the Eurodollar market has been a credit "extender" more than an "allocator." It has extended world credit substantially further than would have been done in its absence. Other so-called "Eurocurrency" markets have had similar effects, but on a smaller scale.

Among the authorities who have concluded that the Eurodollar market has not been seriously expansionary or inflationary based on credit rather than merely monetary analysis, Niehans and Hewson [1976] have probably been the most influential. They find that the Eurodollar market cannot be analyzed as a "money supply" problem, with "fixed coefficient multipliers," but must be viewed as part of a "world credit network" allocating credit through financial intermediation. Although not stating it explicitly, they strongly imply that the total volume of credit to be allocated is more or less fixed. Because of lower transactions costs and other similar factors, the Eurodollar banks have provided a growing share of dollar credit, but, it is implied, this has been mainly at the expense of decreased credit extended to non-financial sectors by US resident banks.

Whether the Eurodollar market has been expansionary they see as depending mainly upon the degree of "net liquidity" created by Eurodollar banks, which depends on "maturity transformation" or the relative maturity of their liabilities and claims. Their examination of the statistics of UK-based Eurobanks for September 1973 indicated that maturity transformation was only moderate. Hence they concluded that the effect of the Eurodollar markets upon world liquidity, and thereby on world spending, was slight.

Following this approach, Little [1979] examined the maturity structure of the rapidly growing assets and liabilities over a later period, 1973–1978, and found that maturity transformation had become quite substantial. She found, however, that maturity transformation by Eurobanking was not significantly different from that by US domestic banking (implying perhaps that this ostensible "substitution" was not on balance expansionary). She concluded, in any case (correctly, I believe), that maturity transformation is not a fruitful way of summarizing the impact of Eurobanking on the economy as a whole, including its inflationary influence. She concluded (incorrectly, I think) that "The multiplier model modified to include interest rate leakages remains the most promising approach to that question" (p. 72). On the basis of the very low multipliers found by the empirical studies, she would apparently conclude that the Eurodollar markets have not been significantly inflationary.

Dufey and Giddy [1978], analyzing the Eurodollar markets primarily from the standpoint of credit markets, also conclude that Eurobanking does not much expand total credit or total spending. The Eurodollar market, they write, "grows by substituting for, rather than adding to, domestic credit intermediation. As a result,... the Eurodollar market is not by itself a major cause of global inflation" (p. 109). With regard to international credit flows, including Eurodollar transactions, they write: "any inflow of funds in one country is exactly offset by outflows in one or more other countries" (p. 166). They also dispute the view that Eurodollar credit extension can expand world spending, writing: "To enable the borrower [from a Eurobank] to spend an amount of x in country A for goods and services, a depositor must have foregone spending at least the same amount in country B... or in country A" (p. 166).

Their views are based partly on a partially defective balance sheet analysis of the mechanics of credit flows involving Eurobanking (pp. 158—164). (Consistent sources and uses of funds analysis would have revealed the error.) But more basically their views rest on the assumption that all transfers of funds through (non-bank) financial intermediaries represent "foregone spending" (i.e., saving) on the part of those who make the transfers. This is true neither domestically nor internationally; its falsity is especially patent in a market of such high liquidity and volatility of funds as the Euromarkets.

Another close student of Eurodollar markets who has focused more on their credit operations than their sheerly monetary aspects is Helmut Mayer [1970; 1971]. He found significant expansionary potentialities, but depreciated them, mainly on the grounds that, except in some special circumstances, Euromarket operations did not expand the world "credit base" nor its "degree of utilization." He failed, I believe, to see the full significance in this matter of Liability Management by US banks and Eurobanks. It is this failure, perhaps, that leads him to the generalization that "Mere redeposits of unused funds, whether they are made by banks or non-banks, do not increase the supply of credit available for the financing of economic activity" [1970, p. 17]. In fact it is in good part the lending of "unused" funds — to banks, Eurobanks, and other financial intermediaries — that has fueled the credit explosion of the 1960s and 1970s.

Helmut Mayer recognizes the expansionary potentialities of the reserve-currency role of the dollar, but underestimates its potency in permitting Eurodollar credit to expand without requiring a contraction of credit by US banks. This may well be because, writing in 1970 and 1971, he postulated fixed exchange rates and gold convertibility[1].

[1] In a monograph seen only after this article was completed, Helmut Mayer [1979] very carefully studies the expansionary effects and other disturbing potentialities of the Eurocurrency markets. He recognizes their expansionary tendencies. He explicitly rejects the "100 per cent offset" theory of Dufey and Giddy. On the whole, he takes an agnostic position: "While it can undoubtedly be said that international bank lending has, on balance, had expansionary effects, ... the causal contribution of the Euro-market to such lending is extremely difficult to evaluate" (p. 42). I would also agree with his conclusion that "the monetary impact of the Euro-currency market will be neither constant nor uniform but will vary greatly according to the type and direction of the capital flows to the intermediation of which it contributes" (p. 38).

Although Mayer's monograph deserves more study than I have been able to give it, I nevertheless feel that his approach underestimates the expansionary influence of the Eurodollar market. He bases part of his contention that the Euromarkets cannot have been *very* expansionary on the fact that non-bank Euro-deposits are very small relative to the total of "money" plus "quasi-money" of most developed nations. This, in my view, is an improperly "monetarist" approach to what influences world spending. It would be better to compare credit *flows* — say the relationship between Eurocurrency credits to non-financial sectors and the world total of international credit flows.

There is also, in my view, too close a relationship posited between funds transferred into Euromarkets and "saving," although Mayer clearly sees that they are far from identical. Capital movements, on which he properly lays much stress, seem frequently to be equated with "surpluses" and "deficits" on goods and services account.

Mayer properly views the major influence of Eurobank operations as lying in their borrowing and lending with the non-financial sectors rather than with other banks. But he fails to appreciate fully, I think, the influence of the inter-bank market on the freedom of the banks to lend, via its aid to their Liability Management (which Mayer very aptly terms "liquidity management").

Finally, Mayer seems to conclude that the main "inflationary" danger of the Euromarkets lies in the large "overhang" of volatile Eurobank liabilities. (As he correctly points out, their liquidation would be more likely to be deflationary than inflationary.) I would see their inflationary influence as being exerted gradually over the years as Euro-credit has added appreciably to a world credit expansion which was already excessive.

Many authorities, Mayer among them, have pointed to the contribution the Eurodollar market made to the financing of the huge deficits of oil-importing countries to the OPEC countries. This contribution to financial stability was indeed substantial, but its inflationary implications should be carefully noted. By extending credit to the oil-importing nations, the Eurodollar market helped sustain the sharp rise in prices and costs that OPEC's actions engendered, while maintaining the liquidity, and thus encouraging higher spending, by the surplus OPEC countries. This contribution of the Euromarket was thus not only "anti-deflationary," but also inflationary in tendency.

VIII. Conclusions

1. The contribution of international finance to the worldwide inflation of the latter 1960s and the 1970s was substantial. There was an unusually strong demand for international credit, and innovations in the structure and operations of domestic and international financial intermediaries made international credit availability and extension unusually and persistently high.

2. The financial impetus towards inflation came much more from expanding non-monetary financial intermediation than from increases in world money stocks. This conclusion is supported by the following:

(a) Statistics for OECD countries show generally substantially higher growth rates for total credit variables and for "quasi-moneys" than for narrowly defined "money" over the 1960s and the 1970s[1].

(b) In several analytical cases examined in this paper the expansion of credit is found to provide stimulus to world spending without entailing any increase in the narrowly defined money stock (Tables 1, 5, 6).

(c) In some important cases where money supplies increased along with credit expansion, the significant increase in "money" was found to reside with banks, central banks, or other financial institutions rather than with the non-financial sectors that do the world's spending (Tables 3, 4, 5).

(d) Repeatedly it was found that the credit operations analyzed in this paper permitted, and presumptively enabled, commercial banks and other intermediaries to expand their lending operations.

3. Although the increased liquidity of the world's non-financial sectors could not be considered a major cause of the inflation, several other "liquidity" factors were of great importance:

(a) The Dollar Currency System, later reenforced by the cessation of US dollar gold convertibility[2], insured that the proceeds of dollar lending and spending would ultimately be held in dollar claims. This enhanced "liquidity" encouraged the expansion of credit by both US banks and Eurodollar banks.

[1] Paul Fagan has kindly computed from OECD statistics the compound annual growth rates of "money," "quasi-money," and "total net borrowing of the non-financial sectors" for each of the OECD countries for which comparable data are available. Data for "money" and "quasi-money" cover the years 1963—1979. For the credit variable, the years covered vary slightly among countries. The growth rates are, respectively: Belgium: 6.9%, 20.7%, and 12.9%; Canada: 8.3%, 14.7%, and 19.6%; Finland: 12.2%, 13.0%, and 23.9%; France: 10.8%, 20.1%, and 14.5%; Italy: 16.3%, 15.2%, and 19.8%; Japan: 14.0%, 15.4%, and 14.1%; Netherlands: 9.0%, 12.4%, and 10.5%; Sweden: 10.0%, 9.3%, and 14:9%; the United Kingdom: 8.7%, 13.1%, and 13.5%; West Germany: 7.9%, 11.1%, and 9.0%; the United States: 5.5%, 10.7%, and 12.5%. The growth rates for nominal gross domestic product were in most of these countries much greater than for "money," reflecting the sharp rise in "velocities" that accompanied the credit processes analyzed in this paper.

[2] The freeing of the US dollar from gold convertibility in 1971 was perhaps the final step in the technical freeing of dollar lending from dependence on world monetary reserves. For brevity, the inflationary role of floating exchange rates has not been assessed in this paper. Floating exchanges, it is true, do not insulate even a very large national economy from international influences on their monetary and credit policies; but they do reduce significantly the international constraints on domestic policies and on international borrowing and lending.

(b) Much of these dollar proceeds were ultimately held, by foreign central banks[1] or by others, in either US bank CDs, Federal Funds sold, short-term US Government securities, or Eurodollar deposits. These modes of investment preserved the liquidity of the dollar holdings while increasing the ability of the US banks and Eurobanks to expand their lending.

(c) The Dollar Currency System also added to the liquid loanable funds made available to banks and other financial intermediaries in countries running favorable balances of payments. The most striking example of this was the contribution that unfavorable US balances of payment made to the loan funds available to foreign financial institutions; but this expansionary influence also occurred when US foreign lending provided the finance to permit some foreign countries to run international deficits against others.

(d) Enhanced liquidity was created for financial intermediaries by the growing practice of Liability Management. This enhanced liquidity permitted and encouraged financial intermediaries to be much more aggressive in their credit expansion.

(e) By various new means, including negotiable certificates of deposit, Federal Funds markets, rollover credits, and floating interest rates, the reduction in the liquidity of those who lent to banks and other intermediaries — and in some cases that of the intermediaries themselves — was minimized.

4. In the light of all this, let us consider Professor Johnson's [1972, pp. 55 f.] statement that the "monetarist" explanation of inflation "rest[s] on the fundamental assumption that a highly developed financial system permits a high degree of substitutability of one type of expenditure for another and one source of finance for another, the ultimate constraint on expenditure being the stock of money on which the pyramidal structure of credit creation rests."

[1] In Table 3, it will be recalled, foreign central banks were quite likely to end up with
the demand deposit proceeds of those countries running financial surpluses as a result of
dollar credit expansion. Typically, the central banks invested these funds in US Treasury
bills, other US open market paper, US bank CDs, or Federal Funds. Such actions eased US
credit conditions, either directly or by helping support US bank Liability Management
operations. If, as was sometimes the case, the central banks invested the demand deposits in
the Eurodollar market, this also tended to ease world credit conditions in the manner ex-
plained in Section VI. Only had they held their reserves as Federal Reserve System deposits
(thus draining the reserves of US banks), would the actions of the foreign central banks have
been "neutral" in the sense that the expansion of loanable funds in some credit markets
would have been offset by reduced loanable funds in others.

There can be no doubt that the international financial system that came into being in the post-World War II years provided an unprecedentedly high degree of substitutability among sources of finance. This was shown both in the variety of sources from which non-financial sectors could secure finance, and also, and especially important, in the alternative means by which financial intermediaries could finance their lending. These high substitutabilities were important causes of the inflationary tendencies of the postwar years.

On the other hand, the basic monetarist assumption that "the ultimate constraint [is] the stock of money on which the pyramidal structure of credit creation rests" was progressively falsified by the financial developments of the postwar years. This is evident in the following:

(a) The ultimate constraint of gold reserves upon credit disappeared. It became little more than a nominal constraint on dollar lending under the Dollar Currency System, and it disappeared completely with the abandonment of gold convertibility and fixed exchange rates in the early 1970s.

(b) With the development of Liability Management bank credit ceased to be much constrained by bank reserves.

(c) Bank credit financed by non-demand deposit funds, and credit extended by non-bank intermediaries, became a much larger portion of total credit. This portion of credit is governed not by bank reserves nor a "monetary base," but by the conditions (e.g., relative interest rates, transactions costs, liquidity, safety, and convenience, and the relative aggressiveness of the lenders) that influence the transfer of funds between demand deposit and other forms of bank and non-bank obligations. The explosive growth of the Euromarkets was a prime example of this virtual independence of non-demand deposit credit extension from anything remotely resembling a "monetary base"[1].

[1] It should not be inferred from earlier material that all other students of the Euromarkets disagree with my conclusions regarding them. For example, Heller [1978; 1980] takes a position broadly similar to mine, and forcefully argues that the growth of Eurodollar credit is principally demand-determined. McKenzie [1976; 1981] sees the Euromarkets as dangerously expansionary and seriously independent of national central bank control. Although Einzig [1964; 1977] extolled the contributions of the Eurodollar market to an efficient international credit mechanism, he cited the great increase in the "velocity of circulation of deposits" and the "considerable additional credit expansion" that the market brought about. His original [1964] list of 32 "changes" brought about by the market contained as many damaging and dangerous as salutory ones. Martin Mayer [1974] recognizes the great expansionary tendencies of the Eurodollar market, and expresses fear that it may be uncontrollable. His unease may reflect the unease of some leading practical bankers, on depth interviews with whom Mayer partly based his insightful book.

5. Indeed the basic paradigm of a "pyramidal" credit system profoundly misrepresents the true nature of the contemporary financial system. The correct paradigm is that of the "pure credit network" suggested by Niehans and Hewson [1976]. In such a network the distribution of credit is governed by a number of factors, among the most important of which are interest rates, non-price terms of lending, information and transactions costs, liquidity, and safety. These factors, as Niehans and Hewson point out, influence the distribution of credit among lenders, intermediaries, borrowers, and markets.

But an important characteristic of the modern credit network, neglected by Niehans-Hewson and by many others, is that the *amount* of credit extended is highly variable, influenced by the above and also other factors, including the growth of money expenditures and receipts and price behavior and expectations. *In "non-convertible" credit networks credit expansion in some parts thereof does not diminish, but instead will probably increase, potential and actual credit extension in other parts of the network.* Such credit networks, in good and especially in inflationary times, have a very strong tendency towards rapid growth[1]. This tendency was manifest in the growth of international and domestic credit during the period examined in this paper.

The ultimate determinants of the behavior of such a credit network are not money stocks but rather: (1) propensities to spend beyond current receipts and to borrow in order to do so; (2) the disposition of financial intermediaries to borrow and lend on attractive terms; and (3) the willingness of so-called "ultimate" lenders (which include, in many cases, financial intermediaries themselves) to provide the funds for the intermediaries to lend[2]. *Interest rates and other terms of lending and borrowing, not amounts of "reserves" or of "money," take center stage in this paradigm.*

[1] Part of this tendency is because of changes in risks and risk-perceptions that accompany booms, and especially inflation. Keynes [1936, p. 145] pointed out that both borrower's and lender's risks constrain spending financed by credit, and observed: "During a boom the popular estimate of the magnitude of these risks, both borrower's risk and lender's risk, is apt to become unusually and imprudently low." During ongoing inflation, of course, the low estimates of these risks may not actually be imprudent.

[2] The most important determinants of the supply of credit in the fully developed financial economy are the modes of operation and the policies of its financial intermediaries. If they characteristically borrow funds in highly liquid forms, they decrease the liquidity of the lenders only very slightly, whereas the borrowers are normally quite willing to accept the reduced liquidity that the debt and debt service entail. More important, the borrowers secure an additional source of funds to spend, without necessarily any corresponding reduction in spending by the lenders. Hence the net result is likely to be "inflationary."

Perhaps the most serious defect of standard contemporary macroeconomic theory is the view that financial intermediaries mediate between deficit units and surplus units between "investors" and "savers" — rather than merely between borrowers and

6. In any case, the standard monetarist contention, namely that it was the expansion of the world's stock of means of payment that caused the worldwide demand-pull inflation of recent years, must be rejected[1]. The standard "Keynesian" explanation is closer to the facts, but must be fleshed out with the effects of profound changes in financial intermediation.

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lenders. (This is perhaps a carry-over from the way the original theory of intermediation was expressed by Gurley and Shaw [1956].) In fact, of course, lenders are not necessarily current savers, and the supply of loan funds through intermediaries is at most very loosely connected with saving. Saving, it is true, must develop *in due course* to match deficit spending. But it is mainly movements in expenditure and income (with possibly some help from interest rates) that equate the two. In inflationary times, the chief danger is not that a high demand for borrowed funds and a shortage of saving will "crowd out" some borrowers; it is that they will *not* do so.

Doctrinally, economists have come to accept Keynes' proposition that surpluses must equal deficits *ex post.* But too many of them ignore the other two major pillars of Keynesian macroeconomics: first, that it is movements of spending and income that equate them; second, that the willingness to lend is determined more by liquidity and safety (and interest rates) than by the propensity to save.

Smith [1978] aptly calls the positive contribution of financial markets to the aggregate levels of money spending and economic activity the "intermediation effect." It is powerful and ubiquitous.

[1] There is perhaps one ultimate "monetary" constraint remaining in credit networks. It will have been noted that a condition for the great expansion of both US bank and Eurobank lending during the inflationary years was their ability to persuade lenders — both non-financial units and other financial institutions — to exchange demand deposit holdings (the major means of payment) for the interest-earning short-term liabilities of the banks and Eurobanks. It is *conceivable* that growing needs for "money balances," as inflation makes world expenditures rise, would make it impossible for the banks and other intermediaries to induce those who lend to them to accept these liquid substitutes at interest rates that would permit the intermediaries in turn to extend credit at interest rates that would not choke off the "excessive" demands for credit. If that should occur, *a limited* monetarist position, namely, that an expansion of the world's "money supply" is a necessary *permissive* condition for the *continuance* of world inflation, would have some limited validity. Whether credit expansion would not be constrained well within these limits by other developments is, of course, another question.

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INFLATION, MONEY SUPPLY AND BUDGET DEFICIT
Theory and an Empirical Exercise

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Introduction

The present paper attempts two things: to trace the discussion on the sources of inflation chronologically across the various schools of thought in economics; and to extend Harberger's (1988) analysis of world inflation based on empirical data to cover the period 1988-2000. A generally accepted theory of inflation is absent in macroeconomics and the actual experiences of inflationary episodes in the Western countries, particularly after 1960 have shaped economic theorizing and model building (Frisch 1983:vii, 1). This is not to undermine the significance of the inflationary experiences before 1960; rather it is to emphasize the onset of the most recent debate on inflation in macroeconomics.

The chronological discussion in Section I follows the treatment of inflation in the Classical and neo-classical schools in their Quantity Theories of Money, the Keynes-Kalecki-Post Keynesian story, the Monetarists and the Structuralist school; followed by exploring the explicit link between budget deficits and inflation. Harberger, in his 1988 essay on world inflation, tests what he terms the 'canonical inflation scenario' in which a fiscal deficit financed via a monetary expansion in the banking system lay at the root of the inflationary process. He finds that in the period 1972-88, inflationary countries siphoned more resources out of their banking systems to finance their public sectors than non-inflationary countries (Harberger 1988:222). In Section II of this paper, this exercise is repeated for the years 1988-2000 where we compare high inflation countries with a control group for finding a relationship between inflation, bank credit and the 'pressure' of the public sector on the banking system.

1.1 Inflation in the Classical and neo-Classical Schools

The Classical school spans the late eighteenth century to the late nineteenth century - comprising Hume, Adam Smith, Ricardo and J.S. Mill. The neo-classical school refers to the works of Walras, Marshall and Pigou - from the late nineteenth century to the middle of the twentieth century (Frisch 1983:218). Their theory of inflation is embodied in their theory of money - the quantity theory of money, which basically postulates a positive relationship between money supply and the general price level. Although the quantity theory plays different roles in the two schools (O'Brien 1975 cited in Frisch 1983:218), it is the similarity in the structure and many assumptions of the equations used in the quantity theory of money that justify combining the two schools in this discussion on inflation.

The most important common characteristic of these two schools, hereafter referred to just as the Classical school, is the separation of the economy into real and nominal sectors. In their model, the quantity of money is irrelevant for determining the employment and output in any economy but determines the price level. There are two versions of this quantity theory of money - a macroeconomic Fisher's Transactions Equation of Exchange (attributed to Irving Fisher) and a microeconomic Cambridge cash balance approach (most closely associated with Marshall and Pigou).

Fisher's Transaction Equation

Fisher's Equation, described the relationship between money supply M, the velocity of money V, the real volume of transactions T and the price level P as:

$$(P)\,(T) = (M)\,(V) \qquad \ldots\ldots\ldots\ldots\ldots\ldots (1)$$

The value of all purchases (*PT*) must equal the existing supply of money (*M*) times the 'average' frequency with which that money changes hands (the velocity of money, *V*). If we assume, as Fisher did, that the velocity (*V*) and the real volume of transactions (*T*) remain constant, then the supply of money (*M*) determines the general price level (*P*). This result holds also because of the assumption that the economy is operating at or very close to full employment. The process is clearer when we understand the determination of output and employment in the 'Classical' economy: A competitive labor market generates equilibrium (full) employment which is associated with a (full employment) output as per the aggregate production function[1] in the economy. Given a price inelastic aggregate supply curve in the goods market (due to assumption of full employment), any increases in supply of money will immediately raise aggregate demand in the goods market, which will be translated into higher prices. This will lower the real wages (nominal wages divided by the price level) which in turn is associated with a reduced supply (excess demand) of labor. As employers compete by increasing nominal wages, real wages are raised once again to their original level, which restores the equilibrium in the labor market. Overall, a monetary expansion will increase the price level and nominal wages but will leave output unaffected.

[1] The aggregate production function expresses real output as a function of the quantities of capital and labor inputs and an index of total factor productivity. Given the existing capital stock, technology and organization of inputs, aggregate output will depend on the quantity of labor input used. Each point on the production function corresponds to a specific combination of real output and labor input.

Cambridge Equation[2]

The Cambridge Equation, as developed by Marshall and Pigou, tries to determine the amount of money households and firms wish to hold to conduct transactions. This is a microeconomic approach to the investment decisions of the economic agent. The division of wealth into money and other financial assets is optimal only when the marginal utility of the last unit of cash held equals the marginal utility derivable from an investment in an alternative asset (Frisch 1983:220).

In this version of the quantity theory, the amount of cash individuals wish to hold is determined by the money demand function

$$M_d = k\,P\,Y \qquad \text{..............(2)}$$

Where M_d is the money demand, k is the fraction of the annual value of national income *(PY)* that people wish to hold. Money supply *(M)* is exogenous, determined by the monetary authorities. Then the condition for monetary equilibrium is the equality of money supply and demand:

$$M = Md \qquad \text{................(3)}$$

Substituting equation (3) into (2), we have

$$M = k\,P\,Y \qquad \text{................(4)}$$

As before, if we assume the constancy of Y (due to the assumptions of full employment and a competitive labor market) and k, then equation (4) implies that money supply (M) determines the price level (P). If the money market is initially in equilibrium, then an increase in the money supply creates disequilibrium in the money market ($M>Md$). Since the values of Y and k are fixed, equilibrium in the money market can be restored only if price level rises. Price levels rise because if households and firms find themselves holding more money than they desire, the excess money balances are used to purchase goods and services. Since the supply of goods and services is constrained by the predetermined full employment level of output, excess demand in the goods market causes the general price level to rise in proportion to the initial increase in money supply.

Thus, according to the Quantity Theory of Money, doubling the money supply would double the price level but leave the relative prices of factors and the volume of output unaffected. This property of money is known as the *neutrality* of money. Having summarized, at the risk of over-simplification, the Classical and neo-classical stance on inflation it is important to point out that the two schools did not agree on every aspect of the Quantity Theory of Money. The Classical

[2] This presentation of the Cambridge equation is based on Snowdon *et al* (1994:56-57).

economists (Hume, Smith, Ricardo and Mill) did not believe completely in the neutrality of money. According to Hume an increase in the supply of money does not directly lead to a rise in prices. The increase in the cash balances of economic agents leads to higher expenditures and in a first round to real effects - namely, to an increase in production and employment. Price adjustment follows only with a time lag (Frisch 1983:226-7). Further, as implied in the possibility of production increases, Hume also did not assume full employment as in the neoclassical school. Lastly, production and employment increases until the general price level is restored to the level corresponding to the increased money supply. Changes in money supply thus induce changes in output and employment as well as in the price level. The neutrality of money is more a characteristic feature of the neo-classical school.

The discussion so far refers to a static economy. Within a growth context, the same may be extended as follows. If growth rate of real output is determined by the natural growth rate (of population and technology), then as long as growth rate of money supply (which is equivalent to the growth rate of bank credit) exceeds the growth rate of real output, there will be a positive growth rate of the price level, i.e., the general price level will continue to rise.

1.2 Keynes-Kalecki-Post Keynesian School

Keynes' explicit discussion of inflation is traced to two sources: Chapter 21 of his *General Theory* and a 1939 essay on *How to Pay for the War*. In the latter, he dwelt upon the relationship between inflation, taxation and the distribution of income that results from increasing expenditures such as war. Keynes' *General Theory* does not subscribe to the Classical assumption of a predetermined full employment real output. In his vision, at levels of output (Y) less than full employment output (Y_F), monetary expansion will increase both output and price levels. Once full employment is achieved, the Classical result of money supply changes affecting only the price level will hold. Inflation is thus a full employment phenomenon.

In Keynes' representation, if the aggregate supply function in the goods market is perfectly inelastic to prices, then monetary expansion will increase output as long as the original equilibrium was at less than the full employment level via changes in the aggregate (effective) demand. However, the effect of money supply on effective demand is not straightforward. It plays indirectly through effects on the liquidity preference, interest rate, the investment and the multiplier. In the Keynesian theory of liquidity preference (the desire to hold cash), the demand for money varies unpredictably causing the velocity of money to vary too. Relating this to the Quantity Theory, if V is variable, then the effect of M on P depends on the direction and magnitude of variation of V. Further Y is also not a constant full employment output in the Keynesian model, with the result money is not neutral and changes in money supply do not lead to directly proportional changes in the price level.

A brief summary of the Keynesian model of output and employment is in order. The principle of effective demand holds that the level of output (hence employment) is determined by the sum of the aggregate planned expenditures by households, firms and the government. Consumption spending by households is endogenous and depends on household income; every household has a marginal propensity to consume. Investment expenditure by firms depends on the expected profitability - the marginal efficiency of capital. Both consumption and investment are not functions of the interest rate, as the neoclassical school holds. Any change in investment expenditure will have amplified effects on output; the multiplier (whose value depends on the marginal propensity to consume) ensures that income (output) changes by a multiple of the change in investment expenditure. All this is based on the assumption that there is spare capacity in the economy; full employment is a special case. In Keynes' theory interest rate was a purely monetary phenomenon, determined by the demand (liquidity preference) and supply of money (determined by monetary authorities). Since liquidity preference is not stable, this does not impart stability to the money demand function (as in the Classical Quantity Theory) and in turn causes the velocity of money also to be unstable. An increase in money supply, by reducing the rate of interest can stimulate effective demand through investment increases and affect output (and employment) through the multiplier effects. While monetary policy received no great role, government fiscal policies (taxation and deficit spending) had a great role in affecting consumer spending thereby affecting effective demand. Keynesian economics thus brought in the role of unstable expectations as it related although indirectly, to inflation.

The role of government is pursued actively in the model of the 'inflationary gap'. In Keynes' own illustration[3], if in a fully employed economy:

the level of output (X) was 5,500 million pounds at prices before inflation,
private income (E) was 6,000 million pounds,
direct taxes (D) were 1,400 million pounds,
private disposable income (E-D) was 4,600 million pounds,
private saving was 700 million pounds and
the government (G) could buy goods and services worth 2,250 million pounds,

then,

[3] As cited in Frisch (1983:228-9).

effective aggregate demand was 4,600 - 700 = 3,900 million pounds,

the volume of goods and services available for private consumption was 3,250 million pounds (X - G) and

the inflationary gap would be 3,900 - 3,250 = 650 million pounds.

The inflationary gap is the excess demand in the market for consumption of goods and services. This would increase the general price level by 20 percent. Real output of 3,250 million pounds at the old prices, would be sold at current prices for 3,900 million pounds (a 20% increase in value) which would lead to unanticipated profits of 650 million pounds which subsequently induces excess demand in the goods market and to further inflationary effects. A spiral results when this excess demand in the goods market pushes up demand in the labor market. Combined with a favorable profit scenario for firms, this bids up money wages to the full extent of the price level. Back at the original level, real wages will create a new inflationary gap which will again increase prices. If money wages follow suit in the next round, this will stimulate further price increases causing a spiral (Frisch 1983:229). In the model, price increases are always ahead of wage and other cost increases and inflation acts like a pump that transfers income from wage earners (with a low propensity to save) to the entrepreneurial sector (with a higher propensity to save). The inflationary process is a redistribution process and the gap is financed when real wages are depressed through a lagged adjustment of money wages.

An important interpretation of Keynes is developed by the Post-Keynesian school. Keynes provides the monetary perspectives while Kalecki provides the real analysis and Sraffa the value and distribution components (Snowdon et al. 1994:367). While regarded as a disparate bunch of economists, certain common themes run through the Post-Keynesians. They focus on the investment component of effective demand; fluctuations in investment explain much of the inherent dynamic of capitalism and are stimulated by changes in long term expectations and financial conditions. They also attempt to establish a firm microeconomic base which has a large degree of correspondence with reality (*ibid* p.370). This encompasses their interest in administered pricing, where firms price goods as a mark-up over costs, the mark-ups being determined by the monopoly powers the firms exercise in the market. This is related to their views on inflation by a macro model of the mark-up. In Post-Keynesian models, inflation is seen to be the result of conflict over the distribution of income with trade union power strongest at times of full employment. Demand-management policies to control inflation have a direct impact on output and employment levels, while intensifying the conflict by altering the amount of money available for distribution. For example, a tight monetary policy of the Government would intensify unemployment; in turn it would be able to induce breaks in the rise of wages and salaries, thereby affecting inflation. They also believe in the endogeneity of money supply; it is dependent on the needs of trade. The banking system is in the active position of accommodating or frustrating private sector expenditure plans. There is an income-generating finance process whereby planned expenditures are transformed via bank credit into actual expenditures. Post-Keynesians effectively hold that the causation runs from commercial bank loans to the money stock to the monetary base. Consequently they believe that the growth in money stock is a response to, not a cause of inflation. This leads to their belief in incomes policies to achieve full employment, not monetary policy.

1.3 The Monetarist School

Orthodox monetarism, of the type christened Mark I, is traced from the quantity theory of money approach as espoused in the 1950s to the mid 1960s (Friedman); through the expectations-augmented Phillips curve analysis (Friedman) which was absorbed into monetarist analysis after the mid-to-late 1960s; finally to the monetary approach to balance of payments theory and exchange rate determination (Johnson, Frenkel) (Snowdon et al. (1994:139).

Friedman presented a restatement of the quantity theory of money not as a determinant of price level but rather as a theory of the demand for money. In his version, the demand for real money balances depends positively on the permanent income of the holder and negatively on the return on other financial assets and the expected rate of inflation. Utility maximizing asset holders will reallocate wealth between different assets, including cash balances, until their marginal rates of return are equal. If the demand for money is stable, then the velocity would be stable. In this case, any observed instability could be attributed to fluctuations in the money supply. Empirical support to this theorizing was given by Friedman and Schwartz (1963) where they showed the stock of money played a largely independent role in cyclical fluctuations in the United States in the period 1867-1960. They held that monetary changes were the cause of major recessions. A major belief of the monetarist school was that money supply should be allowed to grow at a fixed rate in line with the underlying growth of output to ensure long-term price stability.

The quantity theory of money in the monetarist school is different from that in the neoclassical model. In the former, it is a theory of nominal income that maintains that changes in the money stock are the dominant determinant of changes in money income (Mayer 1978:5). Changes in the supply of money alter both the general price level and real variables in the short run; money is not neutral in the short run. However in the long run after all adjustments have taken place, the neoclassical dichotomy between nominal and real sectors is again valid and the results are identical with the neoclassical Quantity Theory of Money.

The second stage of the monetarist school was the expectation-augmented Phillips Curve. A short note on the original Phillips Curve serves as a logical start. In 1958 Phillips derived a statistical inverse, non-linear relationship between unemployment and the rate of change of money wages in the UK for the period 1861-1957. At an unemployment rate of 5.5 percent the rate of change of money wages was zero percent and at an unemployment level of approximately 2.5 percent the rate of change of money wages was 2 percent. Following this work, Keynesian economists derived a similar relationship between unemployment and the inflation rate. The Phillips curve allowed the Keynesian theory of output and employment determination to be linked to the theory of wage and price inflation (Snowdon 1994:150). The mark-up pricing hypothesis they used suggested that price inflation depended on money wage inflation minus productivity growth. They found that at an unemployment level of about 2.5 percent stable prices would prevail because at this level of unemployment, the rate of change of money wages was approximately equal to the then average growth of productivity of 2 percent (*ibid.*). The Phillips curve offered policy makers with a wide menu of choices of unemployment and inflation rate combinations. However, by the late 1960s both inflation and unemployment began to rise and the relationship was challenged by Phelps and Friedman who denied the existence of a permanent (long run) Phillips Curve in their version of the expectation augmented Phillips Curve.

According to them, although money wages are set through negotiations, what matters to the parties is the real wage, which in turn depends on their expectations concerning inflation. Friedman argued that the Phillips curve should have seen the relationship between unemployment and rate of change of real wages instead. He augmented the Phillips Curve so that the rate of money wage increase is equal to the level of unemployment plus the expected rate of inflation. This implies a family of Phillips Curves each associated with a specific expectation of inflation such that the actual relationship is perfectly inelastic and yields a vertical Phillips Curve in the long run (with adjustments) with a natural rate of unemployment. A monetary expansion would increase aggregate demand and this excess demand in goods and labor markets would put an upward pressure on prices and money wages. A money illusion arises - when labor interprets the increase in money wages to be increases in real wages - and would supply more labor. Real wages actually would have fallen, and as firms demanded more labor, unemployment would fall with money wages rising at a higher rate. As workers slowly adapt to their inflation expectations, they demand more money wages at the expected rate of inflation. This would cause firms to layoff workers as real wages rose and unemployment would increase till the original level is restored. Thus in the long run the rate of change of money wages equals the rate of inflation so that real wage is constant. At the natural rate of unemployment, the actual and expected rates of inflation are equal so that in the long run money is still neutral while in the short run it is not. In the long run equilibrium of an economy, at the natural rate of unemployment, the rate of monetary expansion will determine the rate of inflation (assuming a constant growth of output and velocity) in line with the quantity theory of money approach (Snowdon et al. 1994:159).

Friedman's most famous assertion is 'inflation is always and everywhere a monetary phenomenon in the sense that it can be produced only by a more rapid increase in the quantity of money than in output'. Reducing rate of monetary expansion results in an increase in the level of unemployment. Demand management policies can only effect changes in output and employment in the short run. Hence the adoption of monetary rules would be the best way to deal with instability in the economy.

1.4 The Structuralist School[4]

Beginning with the work of Streeten in 1962, structural theories address price changes in the long run in the context of structural changes in the organization of production in the economy. The long run inflationary tendency is traced to the interaction of four factors, which are partly technological and partly behavioral (Frisch 1983:153), and constrict the operation of the market mechanism: differences in productivity in the industrial and service sectors, a uniform rate of growth of money wages in both sectors, different price and income elasticities for the output of the industrial and service sectors and limited flexibility of prices and wages (prices and wages are rigid in a downward direction).

It is clear that inflation is studied in a much broader perspective than in the monetarist school. Structural models assume economic activity can be aggregated into a progressive, industrial sector and a conservative, service sector. Given different rates of productivity growth in the two sectors, a uniform rate of growth of money wages throughout the economy must lead to permanent cost pressures in the service sector which is assumed to have the lower productivity. Because firms in this sector use mark-up pricing, this cost pressure creates cost-push inflation for the economy as a whole. Structural inflation, therefore, implies a change in the relative supply prices of the two sectors. The supply price of the service sector rises relative to the supply price of the industrial sector. This type of inflation also assumes a small price elasticity of demand but a large income elasticity of demand for the output of the service sector.

[4] This section is based on Frisch (1983) Chapter 5.

The cost pressures are part of a social conflict over values of inputs such as the nominal wage and exchange rates and rules for contract indexation which in turn force up the price level (Taylor 1991). Taylor concludes that if indexation dominates, price increases may become inertial, in the sense that real wage targets, for example, are satisfied while at the same time last period's overall inflation is an unbiased predictor of current price growth. If real wages are below the target, money wage claims (via mark ups and prices) can push inflation upwards. In this structuralist model, reduced demand will not strongly affect inflation because it will just lead to output declines in the short run.

1.5 Inflation and the Budget Deficit

A government can finance its deficit either by borrowing or by creating money. By borrowing we mean issue bonds to economic agents and absorb purchasing power. Creating money is actually done by the Central Bank, to whom the government sells bonds in exchange for currency. This process of money creation is given the name *debt monetization*; the government can get funds to finance any amount of deficit in this manner. There are constraints to borrowing from economic agents. Potential lenders (households, foreign sources, domestic institutions) may be unwilling to buy government bonds. Or the size of the deficit may be just too big for the purchasing power with the agents. In which case, money creation could always be relied on. Every time there is money creation to finance a deficit, it adds to the purchasing power available to the economic agents. The government creates enough money such that revenues from money creation, called seignorage (computed as the product of money growth and real money balances[5]), should cover the real deficit (nominal deficit divided by the price level). Thus the smaller real money balances, the higher the required rate of money growth and thus in turn the higher the rate of inflation required to generate a given amount of seignorage (Blanchard 1997: 444). Thus, as money growth increases, so does inflation.

Hyperinflations are typically characterized by increasing inflation rates. Blanchard (1997:444) offers two reasons for this. Firstly, the higher money growth leads to higher inflation inducing people to reduce real money balances, requiring even higher money growth (and thus leading to even higher inflation) to finance the budget deficit. The second reason is that higher inflation often decreases tax revenues and increases the deficit, which in turn requires higher money growth and thus even higher inflation. During hyperinflations, people find ways of reducing their real balances because as inflation increases, the opportunity cost of keeping money increases.

[5] Real money balance is the nominal money stock divided by the price level. The higher the real money balances held in the economy, the larger the amount of seignorage corresponding to a given rate of money growth.

This is done via resorting to barter, frequent wage payments, using foreign currencies as stores of value (seen in dollarization of domestic transactions in many countries of eastern Europe and former Soviet Union in the 1990s). Measures to end hyperinflations therefore focus on stemming money creation along with fiscal measures to reduce the deficit. Wage and price controls are another way of combating hyperinflations, but their success depends a lot on the expectations of the economic agents about the future.

This much said, it is now time to turn our attention to the empirical section of this paper, devoted to replicating Harberger's (1988) analysis of inflation and computation of the pressure of government on the economy.

2.1 An Empirical Exercise

Harberger's (1988) analysis

Harberger's key hypothesis was what he termed the 'canonical inflation scenario' in which a fiscal deficit lay at the root of the inflationary process. The deficit is linked to inflation via monetary expansion when the government turns to the banking system to finance its deficit. He uses two measures to represent the 'pressure' that is put on the banking system by the need to finance the government for different countries for which data were available, in the period 1970-88:

β = (Δ banking system credit to the public sector) $\div$ GDP.
γ = (banking system credit to the public sector) $\div$ total bank credit.

In calculating β, the increase in the banking credit to the public sector over the course of a year is expressed as a fraction of the nominal GDP and is thus the fraction of total GDP that the public sector siphons out of the banking system. It is a flow variable reflecting concurrent action by the public sector (Harberger 1988:218). The data source for computing this variable for each country for the period 1970-1988 was the monetary survey of the International Financial Statistics (IFS, lines 32an, 32b, 32c), published by the International Monetary Fund, various years. The nominal GDP values were also taken from the same source (line 99b) for various countries and years.

The variable γ is a stock variable; it shows the share of public sector in the total bank credit of a country in a particular year. The total bank credit data were taken from line 32 of the monetary survey of the IFS while the public sector claims on the banking credit were taken, as mentioned earlier, from lines 32an, 32b and 32c of the IFS.

In the canonical inflation scenario, the government might draw new credit from the banking system and the banking system in turn curtails this amount from the private sector. This in a way neutralized the effect of β on the money supply. However, as β and γ rise the social costs of curtailing credit from the private sector also rises. It will then be likely that as the government's share in the banking credit increases, there will be a resistance to curtailing private sector credit; with the result the effect of β will not be neutralized any more. Increasing β will then be accompanied by increases in money supply and hence lead to inflationary tendencies.

Harberger computes inflation π as the difference between the Consumer Price Indices of two consecutive years, the source being line 64 of the IFS for various years and countries. He also computes another variable, λ (the percentage increase in bank credit), as follows:

$$\lambda = (\Delta \text{ bank credit during year t}) \div (\text{total bank credit at end of t-1})$$

to avoid what he terms a 'money supply identity'. The volume of credit retains its identity as a policy variable even when the quantity of money is beyond the control of the Central Bank (as in the case of fixed exchange rate countries). Inflation, most often, is driven by the domestic credit component of the banking system. Data for total bank credit were obtained from line 32 of the IFS for various years and countries.

Harberger's first step was to categorize countries based on their inflationary experiences. To be eligible for inclusion, the country had to have population greater than 2 million (to leave out very small nations) and have data on lines 32, 32an, 32b, 32c, 64 and 99b of the IFS for as many number of years as possible.

Countries experiencing a rate of inflation, π, greater than 80 percent a year for two or more consecutive years were classified as *acute inflation* countries. Those countries with inflation rates greater than 20 percent for four or more consecutive years were classified as *chronic inflation* countries. A *control* group of countries passed through the following filters: an inflation rate less than 10.5 percent for the 1970-82 period, and minimum growth rate of 3 percent of real GDP. Countries that survived this filtering were checked for exchange rate experiences and the control group consisted of stable exchange countries with either fixed exchange rates or very slight appreciation or depreciation in the 1970-82 period.

When Harberger compared the values of β, γ, and λ across the three categories of countries, he found that typically they were higher for inflationary countries than the control group. Banking systems in inflationary countries siphoned off more resources for their public sectors; this was also greater for the acute inflation countries when compared to chronic inflation countries. His general conclusion was that fiscal deficits financed at the banking system, high government claims on the available supply of bank credit, and general lack of credit restraint on the part of monetary authorities play an important role in causing the major inflationary episodes in the real world. These results matched the results of his earlier study of the inflationary experiences of the world for the period 1952-1976 (Harberger 1981).

Harberger notes that there were countries that slipped through all the three categories above, but did suffer from inflation. Such countries were picked up by a fourth category, the *devaluation crises*, those with exchange rate crises, which bore a similarity with chronic and acute inflation countries and exhibited certain risky policy behavior like low international reserves, even in periods of no crisis. Another aspect addressed by the study is on major disinflations - the data provided evidence that GDP growth in these countries was higher on the way down from an inflationary peak than it was on the way up. This is a repudiation of the conventional trade-off between inflation and growth. However, he concludes that the evidence is not sufficient because of the sample size of five.

The study relating inflation to government pressure, however, does only point out the coexistence of inflation and the pressure of fiscal deficits and a monetary expansion. It assumes the causality runs from high growth rates of money supply financing fiscal deficits to high inflation.

Replicating Harberger (1988) for 1988-2000

Following the method used by Harberger, IFS statistics were used to compute π, β, γ, and λ for qualifying countries (those with population greater than two million and data availability) for the relevant inflationary years of the 1988-2000 period (see Table 1). This exercise does not deal with devaluation crises and the costs of disinflation which were included in Harberger (1988). The data sources are:

- Lines 32an, 32b, 32c and 99b for computing β (change in public sector credit as a percent of GDP),
- Line 32 for computing λ (percent increase in total bank credit),
- Lines 32, 32an, 32b, 32c for computing γ (the share of public sector credit in total bank credit).

There was a slight change in the way the inflation rate π was calculated. Harberger defined the differences in the Consumer Price Indices (CPI) of two consecutive years as π; in this paper we use the conventional formula for calculating the inflation rate - π is calculated as:

$$(CPI_t - CPI_{t-1})\,(100) \div CPI_{t-1}$$

Acute and Chronic Inflation Countries

The same cut-offs were used to categorize countries as acute and chronic inflation countries - 80 percent or more inflation for two consecutive years for acute and 20 percent or more inflation for four or more consecutive years for chronic inflation countries. When countries met both these criteria, they were listed as both acute and chronic but for the corresponding time periods. For the period 1988-2000, the acute inflation countries were twelve in number, an increase of 8 countries when compared to Harberger's study (see Appendix 1 for the results of Harberger (1988) study) - Brazil, Bulgaria, Lao PDR, Mongolia, Nicaragua, Peru, Poland, Romania, Russia, Sierra Leone, Turkey and Uruguay. There were 26 chronic inflationary experiences - Colombia, Ecuador, Ghana, Honduras, Hungary, Iran, Jamaica, Kazhakstan, Kenya, Kyrgyz Republic, Malawi, Mexico, Malawi, Myanmar, Paraguay, Peru, Poland, Romania, Sierra Leone, Sudan, Tanzania, Turkey, Uruguay, Venezuela, Zambia and Zimbabwe. The numbers of chronic inflationary countries also were higher than those in the Harberger study - 14. Apart from the possibility of a greater number of inflationary episodes, it reflects two important things: the number of countries has expanded in the 1990s from the break up of the Soviet Union; and data availability has vastly improved when compared to the period before 1988. Five countries feature as both acute and chronic inflationary countries - Peru, Poland, Romania, Sierra Leone and Uruguay. Mexico records acute inflationary episodes in two time periods - 1988-92 and 1994-97. Although the latter time period spans only three years, the high rates of inflation (35, 24 and 21 percent respectively) merited inclusion in the chronic inflationary list.

The data for β, γ, and λ, however, were computed only for the inflationary years also noted in Table 1.

Table 1: Global inflationary experiences 1988-2000.

	Inflationary Years	π (%)	λ (%)	β	γ (%)
Acute Inflation					
Brazil	1990-94	1,623.33	1108.9	0.526	53.5
Bulgaria	1995-97	590.02	186.1	0.308	67.9
Lao	1997-99	238.17	71.4	0.011	21.2
Mongolia	1992-94	177.98	65.5		53.5
Nicaragua	1988-91	4,764.95	2764.0	0.315	65.0
Peru	1988-91	3,371.05	1883.7	0.048	35.9
Poland	1988-90	400.66	203.1	0.155	90.8
Romania	1990-94	192.81	108.4	0.112	100.0
Russia	1997-99	148.69	51.2	0.085	64.0
Sierra Leone	1989-91	106.84	47.0	0.030	75.3
Turkey	1993-98	85.80	110.7	0.050	36.2
Uruguay	1988-91	102.00	74.2	0.078	26.1
Chronic Inflation					
Colombia*	1990-96	23.24	34.6	0.002	5.4
Ecuador	1988-99	44.98	11.3	-0.001	11.1
Ghana	1992-97	27.83	52.4	0.062	77.1
Honduras	1993-97	22.74	20.8	-0.018	7.9
Hungary	1989-96	23.50	11.3	0.066	55.4
Iran	1991-96	28.90	25.2	0.053	54.7
Jamaica	1989-96	26.40	31.1	-0.006	-21.8
Kazakhstan	1994-97	39.30	-2.8	-0.0002	20.3
Kenya	1990-94	29.29	18.4	0.010	41.0
Kyrgyz Republic	1995-99	27.7	13.8	0.017	65.4
Malawi	1991-96	34.66	16.4	0.029	53.1
Mexico	1988-92	21.38	38.6	0.006	39.9
	1994-97	34.40	5.4	0.031	1.3
Mongolia	1994-97	49.3	-25.6	-0.008	28.9
Myanmar	1988-99	25.16	29.2	0.043	71.7
Paraguay	1988-91	26.57	37.2	-0.009	-1.6
Peru	1991-94	48.64	77.9	0.003	13.4
Poland	1990-96	35.07	37.8	0.072	69.4
Romania	1994-99	45.80	71.3	0.026	53.9
Sierra Leone	1991-99	25.07	6.7	0.028	94.5
Sudan	1994-97	68.35	46.7	0.021	69.3
Tanzania	1988-96	27.09	18.3	0.021	66.8
Turkey	1988-93	66.00	70.9	0.040	38.3
Uruguay	1991-97	43.50	36.7	0.015	19.5
Venezuela	1988-99	45.36	27.0	0.014	17.8
Zambia	1994-97	34.19	21.0	0.070	85.3
Zimbabwe	1990-99	22.92	26.9	0.015	32.5
Control Group	1988-00				
Australia		1.88	9.4	0.002	11.1
Bangladesh		6.16	12.3	0.009	24.3
Burkina Faso		2.26	4.0	0.005	-21.5
India		9.0	13.6	0.033	46.7
Jordan		4.47	5.4	-0.016	24.7
Korea		5.74	18.6	-0.001	1.5
Malaysia		3.57	18.0	0.003	1.8
Netherlands^		2.16	5.2	0.001	19.3
Pakistan		9.48	10.3	0.027	46.6
Portugal^		4.94	12.6	-0.009	29.5
Spain^		4.71	7.9	0.007	27.0
Tunisia		3.66	7.9	-0.002	7.7
United States		2.9	5.6	0.004	16.2

* Data are for 1990-96 ^ Data are for 1988-1998.

Source: Compiled by author.

Note: Data missing for β for Mongolia 1992-92, Burkina Faso 1988-93; data missing for γ for Myanmar for 1991-93

Control group countries

The control group, as defined by the Harberger study, for the period 1988-2000 consisted of thirteen countries - Australia, Bangladesh, Burkina Faso, India, Jordan, Korea, Malaysia, Netherlands, Pakistan, Portugal, Spain, Tunisia and United States. These countries had a minimum of 3 percent real GDP growth rate in the period 1988-2000 and inflation rates less than 10 percent in the same period. Data on real GDP and inflation rates were taken from two separate sources - for OECD countries, they were available in the OECD Economic Development Outlook 2001 and for non-OECD countries, they were taken from IMF World Economic Outlook 2003. Available data on growth rates for non-OECD countries were averaged for the period 1985-94 and listed on a per year basis for 1995 onwards in the IMF World Economic Outlook. So we averaged the individual growth rates for the period 1995-2000 and ensured that the control group countries met the 3 percent real GDP and under 10 percent inflation rates in both the 1985-94 as well as the 1995-2000 time periods. For OECD countries the data available in the OECD economic outlook were on an annual basis from 1982 onwards. So we averaged the annual growth rates for the1985-2000 period (to arrive at the same time periods for both OECD and non-OECD countries) and control group countries had to meet the growth rate criteria for the period 1985-2000. Seven OECD countries (Australia, Ireland, Korea, Netherlands, Portugal, Spain, United States) and eight non OECD countries (Bangladesh, Burkina Faso, India, Jordan, Malaysia, Oman, Pakistan, Tunisia) survived this filtering process. However, lack of time series data on the other variables used in the study eliminated two (Ireland and Oman) from the control group list.

When we looked at the exchange rate experiences of these countries, only Jordan maintained fixed exchange rates vis-à-vis the US Dollar and only Netherlands and Spain experienced an appreciation of the exchange rates in the period 1988-2000. All the others (except US) experienced depreciation vis-à-vis the United States dollar.

For the control group, π, β, γ and λ values were computed for the entire time period 1988-2000 (except for the Euro countries as noted below). The formulae used are the same as those found in Harberger (1988) and described in the previous paragraphs.

A note on the Euro area

Among the countries in Table 1, only Portugal, Spain and Netherlands are part of the European Union which adopted the euro as the common currency on January 1, 1999. For these countries, the change in currency units posed a problem for calculating the change in banking credit for the year 1998-1999, in the absence of exchange rates information for the local currencies in terms of the euro. This problem was resolved by ignoring that year and 1999-2000 (to avoid break of continuity) for these three countries.

Summary results

A look at Table 1 and Figures 1, 2 and 3 (Figures in Appendix 2) show that the results confirm those obtained by Harberger (1988). Acute inflationary episodes range from high two digit levels (in Turkey) to mid three digit levels in Asia and Eastern Europe all the way to mid four digit levels in Latin America. Chronic inflationary episodes have been more common in the period 1988-2000 and outnumber the stable, low inflation countries.

The public sector accounts for a greater share of the increase in bank credit in inflationary countries; for the control group Figure 1 shows that β is clustered around zero. This is also true for the chronic inflation countries although the spread of the scatter is slightly larger; for acute inflation countries the share of bank credit changes going to the public sector varies from close to zero to 53 percent.

The most stable pattern is evident in the relationship between inflation rates and the percentage increases in bank credit (λ). Figure 2 shows the 45° line, which represents the points where bank credit grows at the same rate as inflation. It is very clear that all the control group countries have seen bank credit growth rates greater than that of inflation while the converse is true of acute inflation countries. Going back to our budget deficit discussion, it is probable that the process of 'neutralizing' effects of β on money supply has occurred in the control group countries, i.e., greater rates of public sector credit growth have probably led to curtailing credit for the private sector. This needs to be investigated some more since the γ levels have also been high for these countries, which, one would expect would lead to increases in the money supply. The chronic inflation countries display a mixed trend; with the number of countries lying above the 45° line slightly more than those lying below it. For the acute inflation countries, there is a clear indication that their inflation rates have grown faster than the growth of bank credit. A more significant observation is that λ rates for acute inflation countries are distinctly higher when compared to chronic inflation and control group countries.

The most inconclusive relationship is between π and γ (Figure 3). However, while all three groups see high shares of public sector in bank credit, acute inflation countries without exception lie in the high γ range. Poland and Romania are special cases; their inflationary years were before the market reform began (see Table 1).

On the whole, we can conclude that high inflation has been associated with a higher rate of growth of bank credit. Next, it has been associated with increases in public sector credit as a fraction of GDP. To a lesser extent, it has been associated with high shares of public sector in total bank credit. The empirical exercise assumes that the causation runs from money supply to inflation; more data and an econometric analysis are needed to test the direction of causation.

3.1 Conclusion

Inflation has been a hotly debated issue in the history of economic thought. The assumptions and beliefs of the various schools of thought about the determination of output and employment have an important bearing on their views about the sources of inflation in any economy. There is a common understanding that high rates of money supply growth and budget deficits in modern economies are associated with high inflation. The important differences in the approaches to the theory of inflation deal with assumptions regarding the endogeneity or exogeneity of money; and the direction of causation between money supply and inflation.

The present paper dealt with locating the theoretical sourcing of the problem of inflation in the various schools of thought. To the Classical and neo-classical economists, it was due to increasing money supply. To the Keynesians it was a full employment, money supply issue. To the monetarists it was solely a monetary phenomenon. To the structuralists it was sourced in the labor market.

Various empirical exercises confirm the relationship between money supply and inflation. Harberger (1988) tested the relationship between inflation, bank credit and budgetary deficits. The study gave ample evidence of the pressure put on the banking system by the public sector. The present paper extended this exercise to cover the period since the publication of Harberger's study and arrived at similar results. The most convincing link was between inflation rate and the growth of bank credit. Specifically, for policy makers, the lesson would be that running big deficits financed at the banking system and letting the public sector absorb too high a fraction of bank credit or allowing bank credit to expand too rapidly were risky policies leading to inflationary consequences.

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Appendix 1

Harberger (1988) Results

Variable		π	β	γ	λ
Acute Inflation cases					
Argentina	(1982-84)	2.546	.125	.325	3.123
Bolivia	(1982-84)	2.690	.142	.600	3.432
Brazil	(1980-85)	1.230	.015	.283	1.421
Israel	(1980-85)	1.383	.275	.363	1.631
Chronic Inflation cases					
Iceland	1980-85	.499	.009	.053	.671
Ghana	1980-85	.500	.047	.894	.549
Sudan	1979-84	.281	.045	.656	.222
Tanzania	1980-84	.290	.078	.936	.216
Zaire	1980-84	.421	.062	.806	.471
Greece	1979-85	.203	.073	.408	.219
Portugal	1974-84	.228	.064	.162	.256
Turkey	1977-85	.450	.068	.466	.491
Yugoslavia	1979-84	.360	.003	.057	.282
Chile	1977-85	.351	.005	.282	.474
Mexico	1979-84	.583	.077	.613	.500
Nicaragua	1977-81	.353	.083	.345	.200
Peru	1980-85	.928	.033	.380	.853
Uruguay	1979-85	.531	.016	.140	.652
Control group: "stable" exchange rates (1970-80)					
Dominican Republic		.092	.012	.287	.160
Guatemala		.109	.004	.186	.148
India		.056	.019	.449	.171
Jordan		.111	.032	.240	.293
Kenya		.114	.014	.252	.257
Malawi		.086	.018	.425	.287
Morocco		.083	.031	.517	.177
Panama		.052	.009	.087	.238
Singapore		.032	<0	<0	.201
Thailand		.075	.023	.251	.225
Togo		.076	.005	.035	.282
Tunisia		.053	.003	.140	.146

Source: Harberger (1988: Table 12.1)

Appendix 2

Figure 1: Inflation rate vs. Beta (Change in bank credit to public sector/GDP).

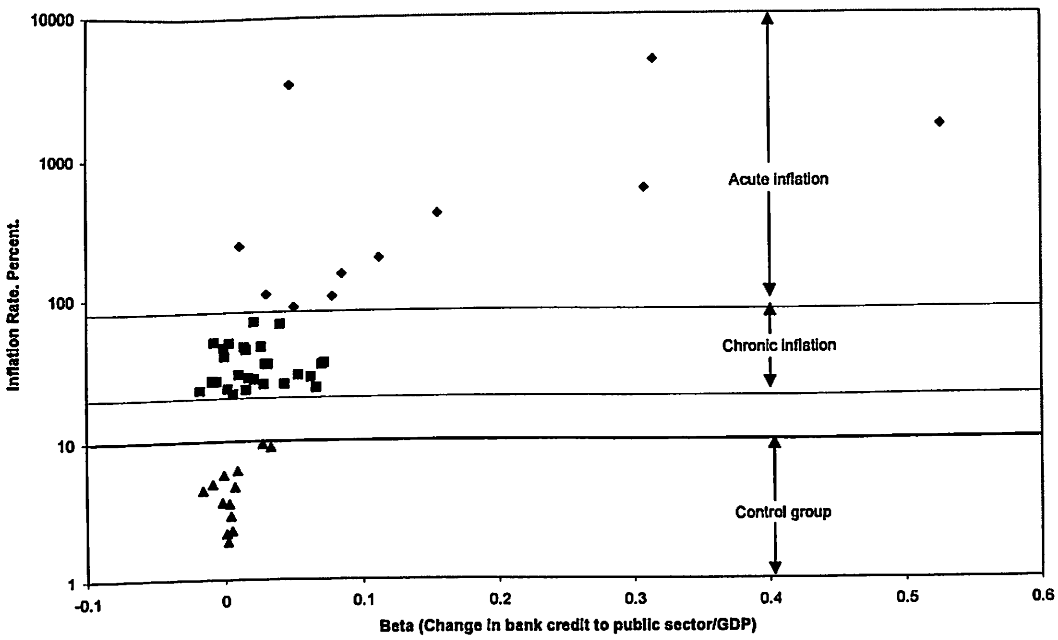


Figure 2: Inflation and Growth of Bank Credit. Percentages.

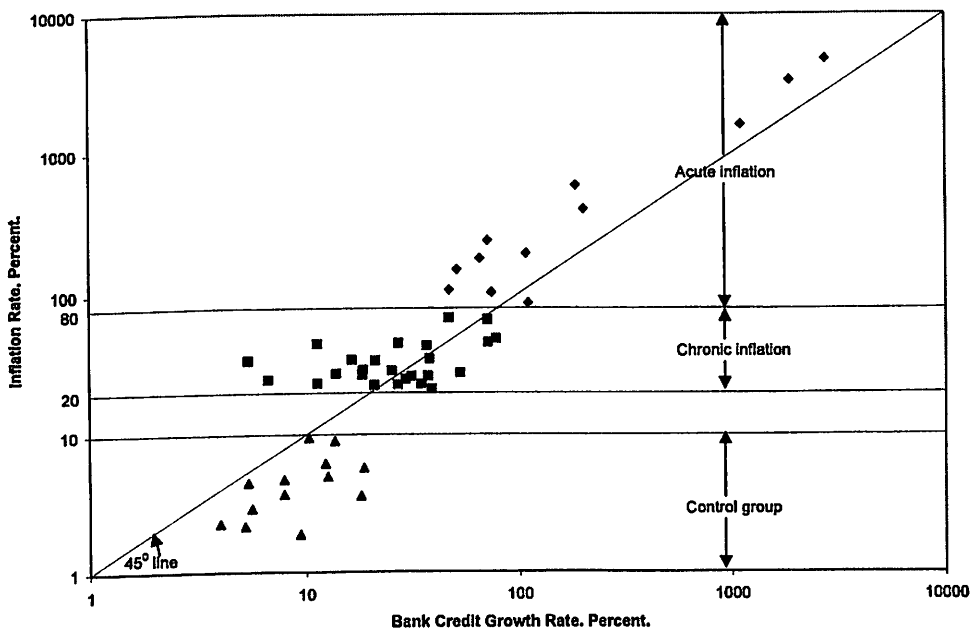
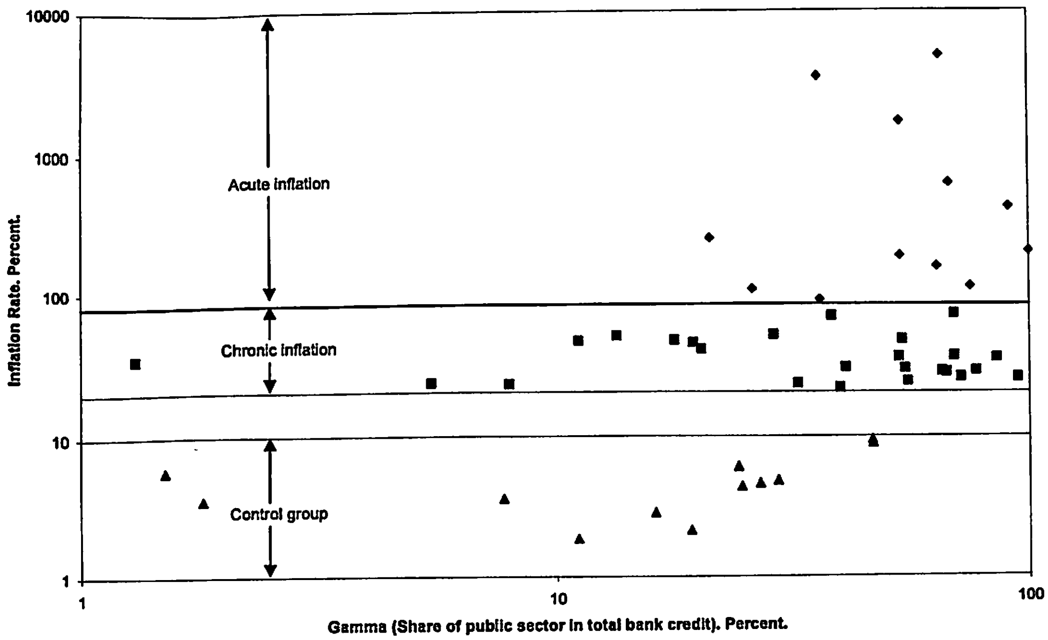


Figure 3: Inflation rate and share of public sector in bank credit.



Note: For a better visual presentation of data in Figures 2 and 3, negative data values in Table 1 have been eliminated.